# Pedalability Index: Cycling Risk — A Street-Level, Injury-Checked Ranking of 300 European Cities

*Victor Rotariu, Defensive Pedal Research · Preprint, October 2026; revised November 2026 (v2.0.3: Stockholm scored on the city's measured traffic counts; v2.0.5: Zurich's Copenhagenize rank corrected)*

## Abstract

**Research Problem and Approach:** City cycling rankings usually measure reputation or policy, weighted by expert judgement, and are rarely tested against injuries per unit of exposure. This paper ranks 300 European cities by the risk of the streets cyclists actually ride, using one per-road risk model that scores about 74 million road segments for a routing service.

**Methodology and Findings:** Scores are aggregated identically into street-network risk, the risk of 210 routed journeys per city, and connected low-stress accessibility, with a 1 km network buffer that corrects administrative-boundary artefacts. Uncertainty comes from 2,000 reruns under varied method choices. Three tiers hold for 246 of the 300 cities; Stockholm, Amsterdam, Helsinki, The Hague and Antwerp lead, and Copenhagen ranks 10th rather than 20th once its boundary stops cutting its bridges. In 18 cities with counter-measured exposure the index predicts serious-injury rates per passage (rate ratio 0.78 [0.70, 0.87] per 10 points), and it correlates with cycling mode share ($\rho = 0.539$, $n = 220$); the injury association disappears when cities are compared only within countries.

**Key Contributions:** The paper provides (1) a single outcome-checked street-level measurement applied across 300 cities, (2) city-level criterion validity reported with and without country effects, and (3) a correction for administrative boundaries that cut networks.

**Implications:** Tiers, not places, are the defensible claim; within-country comparisons need more outcome data. Because the index responds only to physical change, repeated measurement can track whether streets get safer.

**Keywords:** cycling safety, composite index, low-stress connectivity, serious injury, European cities, OpenStreetMap

## 1. Introduction

Which European cities are safe to cycle in? The question sounds simple, and it is asked constantly, by people choosing where to live, by city administrations comparing themselves with their neighbours, and by the media. The answers it usually gets are reputational. The best-known city rankings combine infrastructure inventories, usage statistics and assessments of policy and ambition, weighted by expert judgement. They tell us which cities are admired for cycling. They are not built to tell us how dangerous a city's streets are for the person riding them, and they are rarely tested against that.

Testing a city ranking against injuries is harder than it looks. The obvious yardstick, cyclist casualties per resident, is confounded by exposure. Where more people cycle, each cyclist is less likely to be struck (Jacobsen, 2003), and places where cycling grows tend to become safer per cyclist (Aldred et al., 2017). A city full of cyclists can therefore report many casualties and still be safe, and a city where almost

nobody cycles can report few and be dangerous. Even the choice of denominator changes the answer: distance, trips, population and participation give different verdicts on whether cycling is getting safer (Ferenchak & Marshall, 2020). Without exposure data, a ranking built from casualty counts measures participation as much as danger.

The alternative is to measure the street network directly. Over the past two decades, route-level studies have identified the road characteristics that go with injury: main-road exposure, parked cars, tram tracks, junctions, and the absence of separation from motor traffic (Teschke et al., 2012) (Adams & Aldred, 2020) (Aldred et al., 2021). Crash physics adds speed, since injury probability rises steeply with impact speed (Lubbe et al., 2022). Network research has shown that safe streets matter only if they connect (Furth et al., 2016). Bikeability indices have turned these ideas into maps (Winters et al., 2013) (Hardinghaus et al., 2021) (Krenn et al., 2015). But these indices are almost all built for a single city. Their weights come from experts or surveys. The few that compare cities compare a handful (Kellershohn et al., 2025), and none of those we found has been checked against injury outcomes per unit of exposure.

A street-level measure avoids that trap, at a price. It cannot see culture, enforcement, driver behaviour or the quality of cycling policy, and it does not try to. What it gains is falsifiability. If the index says one group of cities has more dangerous streets than another, injury rates per unit of exposure should be higher there, and where exposure data exist, that can be checked. It also gains a kind of neutrality. Advocacy, plans and announcements do not move a street-level measure; only physical changes to the network do. That makes it useful to anyone who wants to know whether a city's streets are actually changing, not only whether its ambitions are.

This paper describes and evaluates the Pedalability Index: Cycling Risk, a city index for 300 European cities that takes the second route at continental scale. Its starting point is a per-road risk model that a bicycle-routing service uses daily to plan routes across 32 European countries. Every one of about 74 million road segments carries a score. We aggregate those scores identically for 300 cities into three dimensions: - the risk of the street network itself; - the risk of 210 standardised journeys planned by the live router; - how much of a city's population can reach everyday destinations over connected low-stress streets.

Each city's index comes with a rank interval from 2,000 respecifications of the method, following standard practice for composite indicators (Saisana et al., 2005) (Nardo et al., 2005). Cities are grouped into three tiers. The paper then asks whether the result deserves trust, using two checks fixed before the results were seen. The first is whether the index predicts serious-injury rates per counter-measured passage in the cities where both are available. The second is whether it correlates with cycling's share of commuting.

The paper makes three contributions. First, it applies one outcome-checked road-level measurement across 300 cities in 31 countries, so that differences between cities come from their streets rather than from different data or different judges. Second, it reports criterion validity at city level against exposure-controlled serious-injury rates, with and without country fixed effects, and states plainly what the second result does not support. Third, it identifies an artefact that affects any network measure bounded by administrative borders: Copenhagen's boundary cut its bridges out of the network. It corrects the artefact for all cities with a 1 km network buffer and quantifies the effect.

The rest of the paper is organised as follows. Section 2 reviews what is known about cycling injury risk, network connectivity and the construction of composite indices. Section 3 describes the frame, the road-

level model, the three dimensions, the boundary buffer, and the validation design. Section 4 reports the ranking, its uncertainty, the validation results and the sensitivity analyses. Section 5 interprets them, Section 6 sets out the limitations, and Section 7 concludes.

## 2. Background

### 2.1 What makes a street risky for cyclists

The most consistent finding in the injury literature is that separation from fast, heavy motor traffic protects cyclists. A Canadian case-crossover study of 690 injured cyclists compared each injury site with a control site from the same trip. Cycle tracks carried about one ninth of the risk of major streets with parked cars and no infrastructure (adjusted odds ratio 0.11) (Teschke et al., 2012). On major streets, risk was lower without parked cars and with bike lanes. Tram or train tracks tripled it, and downhill grades and construction raised it. In Montreal, with use counts as the denominator, the relative risk of injury on cycle tracks was 0.72 compared with parallel streets, even though the tracks carried 2.5 times as many cyclists (Lusk et al., 2011). In London, also using exposure-controlled case-crossover methods, protected infrastructure reduced injury odds by 40 to 65% in the morning commute. Advisory lanes increased them by 34%, and junctions tripled them (Adams & Aldred, 2020).

Not all studies agree. A British morning-peak study found main roads twice as risky as residential streets, and bus lanes raised injury odds by 84%. It found no protective effect of cycle infrastructure at all (Aldred et al., 2021). A companion analysis also found that lower speed limits lost significance after adjustment, while intersections stayed strongly associated with injury (Kapousizis et al., 2021). Reviews of the field note that the evidence is uneven across facility types (Reynolds et al., 2009) (Mulvaney et al., 2015). A recent synthesis of 104 studies highlights separation, intersection design and signal timing, but reports inconsistent data granularity across studies (Chen et al., 2026).

Speed connects the findings. Injury probability rises steeply with closing speed for unprotected road users (Lubbe et al., 2022) (Rosén & Sander, 2009). A recent review attributes fatality reductions of over 40% to urban 30 km/h limits (Yannis & Michelaraki, 2025).

Two points from this literature shape the index. First, the factors that matter are properties of the street (speed, traffic, road class, separation, parking, junctions), so they can be read from a map. Second, the evidence is strongest for serious injury, and weaker and more contested for crash frequency.

### 2.2 From streets to networks

A safe street helps only if it leads somewhere. When high-stress links are removed, the remaining low-stress network often breaks into islands separated by barriers (Furth et al., 2016). That observation produced the level-of-traffic-stress (LTS) framework. It classifies links into four stress levels, based on a typology of cyclists and on Dutch design standards. Networks are evaluated by the share of origin-destination pairs connected without high stress or excessive detour (Furth et al., 2016). Later work expressed connectivity as bike-accessible jobs and potential bike-to-work trips (Furth et al., 2018) and used it to prioritise new facilities (Lowry et al., 2016). Methods for classifying LTS from different data sources have since been compared (Harvey et al., 2024), and their link to observed behaviour has been tested (Wang et al., 2016).

The reason to care about low stress is that most potential cyclists are not willing to ride in traffic. A national survey of 3,000 adults found that about one third of US adults would not cycle under any

conditions, and examined how the appeal of different facility types varies between types of cyclist (Dill & McNeil, 2016). Longitudinal data from 28 US cities show that protected lanes produced commuter increases 1.8 times larger than standard lanes (Ferenchak & Marshall, 2025). A review of bikeway networks found a consistent positive link between networks and cycling levels, although few studies measure the network as a whole (Buehler & Dill, 2015). Networks also affect risk. In Seville, linking separate bikeways into a network reduced risk beyond what the added length alone would explain (Marqués & Hernández-Herrador, 2017). In the Netherlands, crashes between bicycles and motor vehicles concentrate on distributor roads. A municipality-level study examined whether routing cyclists away from them, a strategy called unbundling, reduces serious casualties (Schepers et al., 2013).

Measuring connectivity raises a geographic problem. Area-based measures, such as street counts per census zone, misrepresent movement because cyclists cross zone edges and zones vary in size and shape (Schön et al., 2024). Reviews of accessibility measures make the same point more generally: almost all have conceptual and computational limitations (Vale et al., 2015). A city boundary is such an area. We found no source that addresses what happens to a city's network score when the boundary cuts its network, which is the problem Section 3.4 takes up.

### 2.3 City indices and how they are built

Bikeability indices are the closest relatives of the index described here. Table 1 compares those in our source pool on four properties: how many cities they cover, what data they use, where their weights come from, and whether they are checked against outcomes. The Vancouver index combines facility availability and quality, connectivity, topography and land use, with weights drawn from surveys and focus groups (Winters et al., 2013). The Graz index derives its components from differences between chosen and shortest routes on 278 trips (Krenn et al., 2015). Two German indices compute their factors from OpenStreetMap, one with weights from an expert survey checked against mode choice (Hardinghaus et al., 2021) (Schmid-Querg et al., 2021). The only inter-city application in the pool covers eight cities and finds bikeability more uniform across them than mode shares are (Kellershohn et al., 2025). A commercial index has been associated with cycling behaviour across 24 cities (Winters et al., 2016). Its validation is against behaviour, not injury.

**Table 1.** Bikeability and city indices in the source pool, compared on scope, data, weighting and outcome checks.

| Index | Cities | Main data | Weights from | Checked against injuries per exposure |
|---|---|---|---|---|
| Vancouver (Winters et al., 2013) | 1 region | GIS layers | Survey, focus groups | No |
| Graz (Krenn et al., 2015) | 1 | GIS, 278 trips | Chosen vs shortest routes | No (behaviour) |
| Munich (Schmid-Querg et al., 2021) | 1 | GIS | Not stated in abstract | No |
| German multifactorial (Hardinghaus et al., 2021) | 1 | OpenStreetMap | Expert survey | No (mode choice) |
| Inter-city model (Kellershohn et al., 2025) | 8 | Programmatic model | Model | No |
| This index | 300 | Per-road risk model, OpenStreetMap, measured and modelled traffic | Outcome-audited road model; fixed dimension weights, respecified 2,000 times | Yes (18 cities) |

The table points to two gaps. Weights in the existing indices are judgements, so the result cannot be falsified: a different panel would produce a different index. And none of them reports whether its score predicts injury per unit of exposure, which is the property a risk index most needs. The literature on composite indicators adds a third consideration. Weighting and aggregation are where construction attracts the most criticism, and robustness analysis is the step most often neglected (Greco et al., 2018). The established remedy is to run uncertainty and sensitivity analysis during construction and to report rankings with their uncertainty (Saisana et al., 2005) (Nardo et al., 2005). Composite indicators of road safety and benchmarks of national road-safety performance face the same issues (Gitelman et al., 2010) (Wegman & Oppe, 2010).

### 2.4 Comparing places by outcome

Outcome comparison between places is the hardest test and the least often attempted. Safety in numbers is consistent from single intersections to whole countries (Jacobsen, 2003). It holds both across British local authorities and over time within them (Aldred et al., 2017), and it is one of three pathways examined in a multilevel study of twelve US cities and over 17,000 fatalities (Marshall & Ferenchak, 2019). Its existence also means that encouraging cycling does not necessarily raise total casualties (Elvik, 2009). Per-km fatality rates show large international differences: in 2018, US cyclist fatality rates per km were four to seven times those of the UK, Germany, the Netherlands and Denmark (Buehler & Pucher, 2020). These comparisons are national. At city level, exposure data are scarce and patchy. Counters exist in some cities and not others (Kraus & Koch, 2021), and crowdsourced traces are a biased sample that must be calibrated against counts (Dadashova et al., 2020). Police records, the usual source of injury data, capture only a fraction of cycling injuries treated in hospital (Macpherson et al., 2024).

The index described here sits where these strands meet. It takes risk factors from the route-level literature, connectivity from the LTS tradition, robustness practice from composite-indicator methodology, and its validation logic from exposure-controlled outcome studies. The open question, which the rest of the paper addresses, is whether these pieces combine into a city ranking that holds up at continental scale.

## 3. Data and methods

### 3.1 The city frame

The frame is 300 cities in 31 countries. It was built by rule rather than by choice. It started from 253 cities covered by the project's earlier work. The remaining 47 places were filled in descending order of population from the Eurostat Urban Audit city-population table, restricted to the 32 countries the routing service covers. Three exclusion rules then removed units that a population table calls a city but a street network does not. A unit was dropped when its scored core municipality held less than half of the population the table lists for it, which is the signature of an agglomeration counted as one city. This rule removed Cannes-Antibes, Mairena, Heerlen, Caserta, Lens, Lucerne, Chester and Bath. In Great Britain, only free-standing units entered the pool: a town that is the namesake core of its own functional urban area, and not part of another city's greater city. This excluded 30 metropolitan boroughs, which would otherwise have entered as cities in their own right. Two Canary Islands cities lie outside the map coverage and were left out. Five cities absent from the Urban Audit (Larnaca, Limassol, Paphos, Strovolos and Reykjavík) stayed in because they were on the starting list.

The resulting frame is uneven by country. Germany contributes 49 cities, the United Kingdom 42, Spain 35, France and Italy 25 each, Romania 20 and Poland 19; ten countries contribute one or two. Romania's share is a legacy of the pilot, which began there, and it weights the lower tail of the distribution. Cities are administrative units, with boundaries taken from the local administrative unit polygons. They hold about 122.9 million residents on the WorldPop population grid (Tatem, 2017), and their street networks total 453,610 km. The median city has 241,736 residents and 1,115 km of streets. Administrative cities are not morphological cities, and a harmonised definition such as the Degree of Urbanisation (Dijkstra et al., 2021) would draw different lines. The administrative unit was kept because it is the unit that builds and maintains streets. Section 3.4 deals with the main way that choice distorts a network measure.

### 3.2 The road-level risk model

The unit of measurement is the road segment. Every public road in the coverage area, about 74 million segments, carries a risk score. The score comes from the per-road model that a bicycle-routing service uses to plan routes, so the index measures the network the router actually navigates. The model reads OpenStreetMap, whose road network is close to complete in Europe (Barrington-Leigh & Millard-Ball, 2017) and reasonably accurate where it has been tested against official mapping (Haklay, 2010). It adds model-side overlays.

The score combines several factor families: - legal and signed speed, including each country's statutory urban default where no sign is mapped; - motor-traffic volume; - lane count and heavy-vehicle access; - the presence and kind of separation from motor traffic, with mandatory-sidepath law respected; - street class and function; - parking-related dooring exposure; - surface; - crossing context.

These are the families that route-level injury studies identify: separation from motor traffic (Teschke et al., 2012) (Lusk et al., 2011), speed (Lubbe et al., 2022), main-road exposure (Aldred et al., 2021), and parked cars and tram tracks (Teschke et al., 2012). The factor weights are proprietary. Following the disclosure policy of the public service, this paper names factors and their direction but not their magnitudes.

Three properties of the model matter for what follows. First, it is identical in every city, so differences between cities come from their streets and not from the scoring. Second, routing and risk data come from the same version of the model, so the router and the scores agree. Third, motor-traffic volume is measured where measurement exists and modelled elsewhere. Measured annual average daily traffic from 43 public sources was matched to 843,935 road segments in 19 countries. Where no count exists, a capacity-constrained gravity assignment model supplies an estimate; where both exist, the measurement wins. A sensitivity variant that uses modelled traffic everywhere is reported in Section 4.5.

The road-level score was validated in a separate programme. It joined permanent, campaign and citizen-science bicycle counters to geocoded injury records: 2,034 counter sites in 13 regions of six countries, 3.98 billion measured passages, and 1,169 killed or seriously injured (KSI) records. Two features of that design matter. Counters measure exposure directly, which avoids the confounding that safety in numbers introduces into per-capita comparisons (Jacobsen, 2003) (Aldred et al., 2017). And the outcome is serious injury rather than all crashes. The programme was first run on an earlier version of the model and was re-run for this paper on the current version, with the same sites, outcomes and specification: a negative binomial model with city fixed effects, log passages as a free exposure term, and errors clustered by region. The current scores predict severity much better than frequency. All

crashes rose by 1.08 [1.00, 1.16] per standard deviation of score, serious injuries by 1.18 [1.12, 1.24]. Serious injury per passage was 1.51 times higher [1.35, 1.68] on roads scoring 80 or more than on roads below, and 1.60 times higher [1.33, 1.92] on roads above 100 than on roads below 30. The current version scores roads higher than the earlier one (20% of sites at 80 or more, against 11%), so fixed thresholds do not compare directly; at the same share of sites the ratio is 1.59 [1.30, 1.94], against 1.62 under the earlier version. With road class controlled, the serious-injury association held (1.15 per standard deviation [1.06, 1.24]). It held on mid-block links (1.27 [1.08, 1.50]) but not at junctions (1.05 [0.87, 1.26]). The result does not depend on the citizen-science counters, which make up about two thirds of the sites. On the 632 sites counted by public authorities alone, serious injuries rose by 1.19 [1.14, 1.24] per standard deviation, and by 1.09 [1.02, 1.16] with road class controlled; on that smaller panel, however, the threshold contrasts lose significance once road class is controlled. Because this panel also helped audit the model's factors, the re-run shows consistency, not an independent test.

### 3.3 Three dimensions

The index combines three dimensions, each answering a different question about a city.

**Street-network risk (S, weight 0.40)** asks how risky the city's streets are on average and in their worst tail. It combines two components. The first is the length-weighted mean comparable risk per street-km (six tenths of S). The second is the share of street length in the highest-risk band (four tenths). The tail is there because the validated severity signal is concentrated at the top of the score, and a city should not be able to hide a few dangerous arterials inside a pleasant average.

Two rules decide which streets count. Roads signed above 80 km/h are excluded as bypasses rather than urban fabric. A carriageway with a parallel protected track, whether detected geometrically or implied by mandatory-sidepath tagging, is scored as the track: a road that cyclists may not use, or would not need to, is not counted as ridden.

**Journey risk (R, weight 0.30)** asks what a typical trip is like. Each city gets 210 standardised trips. Origins are drawn from the population grid and destinations from the distribution of everyday amenities. Each trip is planned by the live production router, which is built on the OSRM engine (Luxen & Vetter, 2011), and scored on the per-road values. R measures the network as experienced rather than as averaged. A city with a coherent low-risk sub-network scores much better on R than its street average suggests, because the router finds that network. A city whose calm streets do not join up cannot hide on R. In total, 62,942 trips were routed. Every city reached 210 except Venice, where water limits routing to 165.

Routed trips are not chosen trips. Cyclists do not necessarily take the objectively safest route (Łukawska et al., 2026), and accident risk and perceived safety pull route choice in different ways (Huber et al., 2024). R is therefore a standardised probe of what the network makes possible, not a forecast of what riders do.

**Low-stress accessibility (A, weight 0.30)** asks how much of daily life a city's calm streets can reach. It follows the low-stress connectivity tradition (Furth et al., 2016) (Furth et al., 2018): once high-stress links are removed, the question is whether what remains still connects people to destinations. A is computed on the 1 km population grid as two components: - the population-weighted share of daily destinations reachable over a connected low-stress network (seven tenths of A); - the share of street length in the largest connected low-stress component (three tenths of A).

"Low-stress" is defined by a threshold on the per-road score. A single cut creates a cliff for cities whose street scores cluster near it, so A is averaged over three thresholds.

### 3.4 The boundary buffer

Administrative boundaries cut through street networks. Computing R and A on the network inside each boundary only causes an error, and it is clearest in Copenhagen. Its municipal polygon has a hole where the independent municipality of Frederiksberg sits, and it leaves out the harbour. The network inside the polygon therefore lacked the bridges and through-streets that join the city's districts. Its largest low-stress component held only 58% of low-stress length, and 9% of routed length crossed streets that went unscored.

The general point is that area-bounded measures misread movement that crosses area edges (Schön et al., 2024). The fix applies to every city. The street network used for R and A now extends 1 km beyond the boundary. Residents, destinations and trip origins stay inside it, and A counts length only for edges whose midpoint lies inside it. Connectivity is therefore assessed on the network a rider can actually use, but credited only to the city that owns the streets. A first version of the fix also counted the length in the buffer ring. That moved Valletta, a small city surrounded by others, 29 places, and the version was discarded.

### 3.5 Normalisation, uncertainty and tiers

Each raw input is normalised against goalposts at the 2.5th and 97.5th percentiles of the 300-city distribution, and the three dimensions are combined with their weights into a 0-100 index. Because the goalposts come from the frame, adding or removing cities moves every index value even when no street changes. Index values are therefore relative to these 300 cities.

Composite rankings are sensitive to weighting and aggregation choices (Greco et al., 2018). Good practice is to make that sensitivity visible rather than assert a single order (Saisana et al., 2005) (Nardo et al., 2005). The index was therefore recomputed 2,000 times, each time with a different plausible choice of weights, goalposts and aggregation (a respecification). Every city is reported with a 90% rank interval. Cities are grouped into three equal-count tiers of 100. A city counts as confidently placed when its modal tier holds in at least two thirds of the respecifications. The paper's headline claims are made at the tier level, not the rank level.

### 3.6 Validation design

Three checks were fixed before the results were computed, and two of them were hard gates. Failing a gate meant stopping and reporting the failure, not adjusting the method until it passed.

**Criterion validity (gate).** For cities that contain counter sites with measured exposure and police-recorded injuries, the KSI rate per passage was regressed on the index in a negative binomial model with log passages as offset, the standard approach for over-dispersed crash counts (Lord & Mannering, 2010). The gate required the 95% interval of the rate ratio per +10 index points to exclude 1. The model was also fitted with country fixed effects, which give each country its own baseline injury rate and so compare each city only with panel cities in the same country. All crashes, not only KSI, were tested as a secondary outcome. Police data under-record cycling injuries, especially those not involving a motor vehicle (Macpherson et al., 2024) (van Petegem et al., 2020). The KSI outcome inherits that limitation. The test is also not fully independent of model development. The counter sites inside indexed cities

come from the same counter programme whose road-level results were used to audit the model's factors, and components that failed that audit were removed before city aggregation.

**Convergent validity (gate).** The index was rank-correlated with the bicycle share of commuting from the Eurostat Urban Audit, with a pre-registered threshold of $\rho \geq 0.45$. Mode share is not risk. It was chosen because low-stress provision and cycling levels are linked in longitudinal multi-city data (Ferenchak & Marshall, 2025) and network reviews (Buehler & Dill, 2015), so a valid risk index should correlate with it moderately, not perfectly. The Urban Audit's mode-share figures come from different survey years in different cities, some of them old, which adds noise to this check.

**Traffic-source sensitivity.** The headline index uses measured traffic where it exists. A variant that uses modelled traffic everywhere was compared with it by rank correlation, tier changes and median rank shift by country. A country whose median shift exceeded ten places was to be referred for a decision rather than corrected silently.

# 4. Results

## 4.1 The ranking

Stockholm ranks first with an index of 98.9. Table 2 shows the top twenty and the bottom ten. The leading group is tight, and its ranks are uncertain. Amsterdam (97.9), Helsinki (97.9), The Hague (97.8) and Antwerp (97.7) all lie within 1.2 points of Stockholm. Each of them reaches the podium in some respecifications, and Stockholm's own 90% interval runs from first to eighth. The top ten is completed by Vantaa, Leiden, Haarlem, Espoo and Copenhagen, which is tenth at 95.0. Dutch and Finnish cities hold eight of the top ten places. Paris (13th), Utrecht (14th) and Barcelona (16th) follow. Among the 47 cities filled from the population table (Section 3.1), the best placed are Nijmegen (18th) and Arnhem (22nd).

**Table 2.** Top twenty and bottom ten cities, with 90% rank intervals from 2,000 respecifications.

| Rank | City | Country | Index | 90% interval |
|---|---|---|---|---|
| 1 | Stockholm | SE | 98.9 | 1-8 |
| 2 | Amsterdam | NL | 97.9 | 2-15 |
| 3 | Helsinki | FI | 97.9 | 1-11 |
| 4 | The Hague | NL | 97.8 | 2-14 |
| 5 | Antwerp | BE | 97.7 | 3-15 |
| 6 | Vantaa | FI | 97.2 | 2-11 |
| 7 | Leiden | NL | 97.1 | 2-12 |
| 8 | Haarlem | NL | 96.6 | 1-14 |
| 9 | Espoo | FI | 95.5 | 3-16 |
| 10 | Copenhagen | DK | 95.0 | 4-22 |
| 11 | Tampere | FI | 94.1 | 4-18 |
| 12 | Eindhoven | NL | 93.8 | 10-26 |
| 13 | Paris | FR | 93.7 | 6-19 |
| 14 | Utrecht | NL | 93.5 | 10-24 |
| 15 | Rotterdam | NL | 93.2 | 9-23 |
| 16 | Barcelona | ES | 93.1 | 8-22 |

| Rank | City | Country | Index | 90% interval |
|---|---|---|---|---|
| 17 | Brussels | BE | 91.6 | 3-26 |
| 18 | Nijmegen | NL | 90.5 | 15-30 |
| 19 | San Sebastián | ES | 90.3 | 18-31 |
| 20 | Turku | FI | 89.6 | 6-26 |
| 291 | Strovolos | CY | 8.8 | 285-294 |
| 292 | Craiova | RO | 7.2 | 290-294 |
| 293 | Galaţi | RO | 6.0 | 291-295 |
| 294 | Paphos | CY | 5.3 | 292-295 |
| 295 | Botoşani | RO | 2.6 | 292-296 |
| 296 | Buzău | RO | 2.0 | 294-298 |
| 297 | Ploieşti | RO | 1.4 | 295-298 |
| 298 | Brăila | RO | 0.6 | 297-299 |
| 299 | Iaşi | RO | 0.0 | 299-300 |
| 300 | Larnaca | CY | 0.0 | 296-300 |

The bottom of the table is concentrated in two countries. Thirteen of the last twenty cities are Romanian and four are Cypriot; the other three are Catania, Athens and Split. Bucharest ranks 286th and Athens 284th. Iaşi and Larnaca both score 0.0. Because the scale is anchored on this frame's own percentiles, a zero marks the bottom of the 300-city distribution, not an absolute absence of safe streets.

Uncertainty is large in the middle of the distribution and small at the ends. The median 90% rank interval is 35 places wide. The extremes are tight: Larnaca, Iaşi and Brăila cannot leave the last five, and Stockholm cannot leave the top eight. A difference of a few places between two mid-table cities carries no information. This is why the paper reports tiers. The tier cut points are 63.0 and 47.9. Members keep their tier in 97% of respecifications in Tier 1, 74% in Tier 2 and 90% in Tier 3. Of the 300 cities, 246 are confidently placed: 98 in Tier 1, 62 in Tier 2 and 86 in Tier 3. The 54 that straddle a boundary include Bergen, Prague, Turin, Glasgow and Saarbrücken.

### 4.2 Countries and the spread within them

Table 3 gives country means for countries with at least four indexed cities. Finland (91.2) and the Netherlands (86.7) lead. Belgium, Denmark, Spain and Sweden follow, all above 69. Romania (14.3) and Cyprus (10.7) are last, below Greece, Portugal and Bulgaria. The United Kingdom (44.3) and Italy (41.3) sit in the lower-middle range, beneath Poland (53.6).

**Table 3.** Index by country, countries with four or more cities.

| Country | Cities | Mean | Median | Rank span |
|---|---|---|---|---|
| FI | 6 | 91.2 | 94.8 | 3-61 |
| NL | 15 | 86.7 | 90.5 | 2-119 |
| BE | 5 | 73.6 | 86.7 | 5-248 |
| DK | 4 | 72.6 | 69.2 | 10-145 |
| ES | 35 | 70.1 | 72.1 | 16-228 |
| SE | 5 | 69.9 | 63.0 | 1-167 |
| NO | 4 | 66.6 | 66.7 | 69-110 |
| FR | 25 | 64.5 | 60.9 | 13-252 |

| Country | Cities | Mean | Median | Rank span |
|---|---|---|---|---|
| DE | 49 | 63.5 | 62.2 | 21-239 |
| PL | 19 | 53.6 | 54.9 | 91-226 |
| CH | 5 | 53.5 | 52.3 | 129-200 |
| GB | 42 | 44.3 | 45.8 | 71-278 |
| CZ | 4 | 44.2 | 43.4 | 176-238 |
| IT | 25 | 41.3 | 40.1 | 45-282 |
| BG | 4 | 32.3 | 33.9 | 230-274 |
| PT | 4 | 28.8 | 31.0 | 247-280 |
| RO | 20 | 14.3 | 14.8 | 246-299 |
| CY | 5 | 10.7 | 8.8 | 271-300 |

Country means hide large spreads within countries. French cities run from 13th to 252nd, German cities from 21st to 239th, Spanish cities from 16th to 228th, and Italian cities from 45th to 282nd. Belgium's five cities span 5th to 248th. Some countries are tightly grouped: Finland's six cities all lie in the top 61 and Romania's twenty all lie below 245th. For most large countries, though, a city's own streets say more than its country does. This matters for interpreting the validation results below.

Spain ranks fifth by mean. Part of this may be legal rather than physical. The model scores each country's statutory urban speed default wherever no sign is mapped, so a country with a low national default lends that default to every untagged street. The model's defaults table, which records the statute for each country, sets Spain's urban default at 30 km/h and 50 km/h only on streets with two or more lanes per direction (Royal Decree 970/2020). The index is therefore sensitive to what the law says as well as to what the streets look like. Section 5 returns to this.

### 4.3 The three dimensions

The dimensions are related but not interchangeable. Oriented so that higher means worse on all three, street risk and access correlate at 0.79, journey risk and access at 0.69, and street risk and journey risk at only 0.46.

The weak link between S and R is the most informative result of the analysis. It means that a city's average street does not predict what a routed trip through it is like. The gap between them, street risk minus journey risk, measures how far a city has a coherent low-risk sub-network that the router can exploit. The largest gaps belong to Reggio Emilia (87.9), Oulu (85.5), Turin (84.6), Milton Keynes (82.3) and Uppsala (81.2). These cities have hostile average streets but a usable network inside them. The smallest gaps, where trips are no safer than the average street, belong to Split, Patras, Reggio Calabria, Ploiești and Messina. In those cities there is no refuge for the router to find. For them the policy question is about connecting the safe streets they have, not about adding more.

Accessibility is the most demanding dimension. At the central low-stress threshold, the median city leaves 63.6% of its residents locked to high-stress streets for most everyday destinations (mean 54.6%). This matches the islands pattern that motivated low-stress connectivity analysis (Furth et al., 2016): calm streets exist in almost every city, but in most they do not join up.

### 4.4 Validation

Table 4 summarises the validation results.

**Table 4.** Validation of the city index.

| Check | Result | Gate |
|---|---|---|
| KSI rate ratio per +10 index points (18 cities) | 0.78 [0.70, 0.87], p<0.001 | Pass (CI excludes 1) |
| Same, modelled traffic everywhere | 0.79 [0.70, 0.88] | Pass |
| Same, public-authority counters only (12 cities) | 0.76 [0.68, 0.86] | Robustness |
| Same, with country fixed effects (5 countries) | 0.95 [0.82, 1.11], p=0.54 | Not a gate; null |
| All crashes per +10 index points | 0.96 [0.88, 1.05], p=0.36 | Not a gate; null |
| Spearman with bicycle commuting share (220 cities) | 0.539 | Pass (threshold 0.45) |

**Criterion validity.** Eighteen indexed cities contain counter sites with measured exposure and police-recorded injuries. Together they cover about 2.17 billion passages and 857 KSI records, in Baden-Württemberg, Belgium, Berlin, Bordeaux, Madrid, Nantes, Paris and Toulouse. In these cities a higher index predicts a lower rate of serious injury per passage. Each additional 10 index points goes with a rate ratio of 0.78 [0.70, 0.87], and the rank correlation between the index and the rate is −0.33. Grouped by tier, the ten Tier 1 cities have 3.41 KSI per 10 million passages and the seven Tier 2 cities 5.43. The highest rates are in Potsdam (7.47), Zurich (7.40), Basel (5.66), Berlin (4.67) and Bordeaux (4.01). Two cities, Liège and Charleroi, recorded no KSI in the counter catchments. Charleroi is the panel's only Tier 3 city, so the panel says nothing about the lowest tier. The result holds when modelled traffic replaces measured traffic (0.79 [0.70, 0.88]). It also holds on the twelve cities left when the citizen-science counters are removed (0.76 [0.68, 0.86]), a step that drops all five Belgian cities and Potsdam.

This result has two limits. First, with country fixed effects the association disappears (0.95 [0.82, 1.11]). Within each of the four countries that have more than one panel city, the ordering of index and injury rate is not consistent. Five countries of outcome data cannot separate a city's streets from the national reporting, riding culture and infrastructure that come with them. The criterion result therefore supports differences between groups of cities, which are largely differences between countries. It does not support fine distinctions within a country. Second, all crashes do not follow the index (0.96 [0.88, 1.05]). This matches the road-level finding that the score tracks severity rather than frequency.

**Convergent validity.** The index correlates with the bicycle share of commuting at $\rho = 0.539$ across the 220 cities that can be matched to the Eurostat Urban Audit. This clears the pre-registered threshold of 0.45.

**Comparison with a reputational index.** The Copenhagenize Index scores policy, culture and ambition as well as infrastructure. Against its 27 European top-30 cities, our ranks correlate at 0.49, and 17 of its 27 cities fall inside our top 50. The two disagree most on Bordeaux, which Copenhagenize ranks 9th and we rank 128th, and Zurich, its 22nd and our 170th. Both cities have injury data in the panel. Bordeaux has the highest KSI rate of the four French panel cities (4.01, against Paris's 1.93), and Zurich has the second-highest rate in the whole panel. In both cases, the outcome data is closer to this index than to the reputational one.

### 4.5 Sensitivity to the traffic source

Replacing measured traffic with modelled traffic everywhere barely changes the ranking. The rank correlation between the two variants is 0.999, the median absolute shift is one place, and eight cities change tier. No country's median shift exceeds ten places. The largest is Iceland at nine, which is a single city (Reykjavík), followed by Great Britain at three.

The cities that move most are large British and German cities, where measured counts show heavier traffic than the model assumed. All of them rank lower with measured traffic: Manchester by 25 places, London by 22, Aachen by 16 and Portsmouth by 14. The headline therefore keeps measured traffic. The cost is that cities with count data are measured more accurately than cities without it, and Section 6 discusses that asymmetry.

### 4.6 Effect of the boundary buffer

The buffer leaves most cities where they were but moves a few substantially. Its overall rank correlation with the unbuffered result is 0.9994 and the median shift is one place, but the 90th percentile shift is four places, the maximum is 24, and four cities change tier. The largest mover is Orléans, a narrow commune whose trips are now scored through its neighbours' streets; it moved from 122nd to 98th. Copenhagen moved from 20th to 10th and Amsterdam from 9th to 2nd.

## 5. Discussion

### 5.1 What the index shows

Measuring the streets rather than the reputation produces a ranking that is familiar at the top and unfamiliar in the middle. The familiar part is that Dutch, Finnish and Danish cities lead, and Amsterdam, Copenhagen, Utrecht and Paris all sit in the top fifteen. The unfamiliar part is that the Helsinki suburbs of Vantaa and Espoo rank sixth and ninth, above most of Europe's famous cycling cities. Index constructions built on reputation rarely evaluate suburbs at all. A street-level measure does not know what a suburb is. It sees calm, connected, separated networks, and the Finnish suburbs have them.

The three dimensions are related but carry different information. Street risk and access move together ($\rho = 0.79$), which makes sense, since access depends on how much of the network is low-stress. Street risk and journey risk correlate only weakly ($\rho = 0.46$). The gap between them identifies two kinds of city. The first, such as Reggio Emilia, Oulu, Turin and Milton Keynes, have a coherent low-risk sub-network inside a hostile average. The second, such as Split, Patras and Reggio Calabria, have nothing for a router to find. This is the network effect the literature describes, from low-stress islands (Furth et al., 2016) to the safety gain of linking bikeways in Seville (Marqués & Hernández-Herrador, 2017), observed here at the scale of 300 cities. For a city administration, the gap is a practical diagnostic. A city with a large gap needs its calm network extended and joined up. A city with a small gap needs one built.

### 5.2 Boundaries are part of the measurement

The Copenhagen case is the most transferable lesson of this paper. Without the buffer, Copenhagen ranks 20th. That result does not reflect its streets. It reflects the outline of its municipality, which leaves out Frederiksberg, an enclave in its middle, and the harbour its bridges cross. Low-stress connectivity measured inside that polygon found a network broken into pieces, because the pieces that joined it lay outside the line. After the 1 km network buffer, Copenhagen ranks 10th and Amsterdam 2nd. The buffer moves most cities by one place or none (median shift one, 90th percentile four), so it is not a re-ranking. It removes an error that fell on a few cities, most of them administratively fragmented.

The general point applies to any city comparison built on network measures. Area-bounded connectivity misreads movement that crosses area edges (Schön et al., 2024). Administrative boundaries are areas drawn for governance, not for travel. Harmonised urban definitions such as the Degree of Urbanisation

(Dijkstra et al., 2021) change which cities are compared, but they do not remove the problem, because any boundary cuts a network somewhere. Our fix keeps the administrative unit for accountability, since it is the unit that builds and maintains streets, while letting the network extend past it. It credits only the length that the city owns, so a small city does not borrow its neighbour's streets. The discarded first version, which counted buffer length and moved Valletta 29 places, shows that crediting matters as much as connecting.

### 5.3 How much the validation supports

The criterion result is the strongest evidence that the index measures something real. In the eighteen cities with counter-measured exposure, a higher index predicts fewer serious injuries per passage (rate ratio 0.78 per +10 points). The association holds with modelled traffic. It is a severity result, as the road-level validation predicted: all crashes do not follow the index. This fits the injury literature, where evidence on what makes streets dangerous is strongest for serious outcomes and most contested for crash counts (Section 2.1). It also means the index should be read as a measure of how badly cyclists get hurt, not of how often they fall off.

The country fixed-effects result limits that claim, and it should not be played down. Once each country's own level is removed, the association disappears (0.95 [0.82, 1.11]). Within each of the four countries with more than one panel city, the ordering is inconsistent. There are two readings, and the available data cannot choose between them. The first is that the index captures differences between national road systems (speed regimes, design standards, traffic levels) and these drive the result, while differences within countries are too small or too noisy to detect with eighteen cities. The second is that national injury reporting differs between countries in ways that happen to line up with the index, so the between-country association is partly artefact. Police data capture only a fraction of cycling injuries (Macpherson et al., 2024), and that fraction plausibly varies by country. The honest conclusion is that the criterion supports the index at the level of tiers and broad groups of cities. It does not support claims that one city in a country is safer than another a few places below it. The rank intervals already discourage those claims, and the validation agrees with them.

The convergent check is easier to interpret. The correlation with mode share ($\rho = 0.539$) is moderate. That is the right size: a risk index that tracked mode share perfectly would be measuring participation, and one unrelated to it would contradict the evidence that low-stress provision raises cycling (Ferenchak & Marshall, 2025) (Buehler & Dill, 2015). Safety in numbers runs both ways here (Jacobsen, 2003). Safe streets attract cyclists, and more cyclists make each one safer. The index measures only the first link, which is why it was designed not to use mode share as an input.

### 5.4 Streets and reputations

The index agrees with the best-known reputational ranking more than it disagrees ($\rho = 0.49$ over 27 cities, with 17 of them in our top 50). Its disagreements are informative. Copenhagenize ranks Bordeaux 9th and Zurich 22nd; this index ranks them 128th and 170th. In both cases the injury panel sides with the street measurement. Bordeaux has the highest serious-injury rate among the French panel cities, and Zurich has the second-highest rate in the panel. Two cases do not make a test. They do show what a reputational index rewards that a street index does not: policy momentum, plans and culture. These may well predict future streets, but they do not describe present ones. The two kinds of ranking answer different questions, and both have uses. The difficulty arises when a reputational ranking is read as a safety ranking.

### 5.5 The East-West gradient and what to do with it

The country table shows a strong gradient. Finland, the Netherlands, Belgium and Denmark sit at the top. Romania and Cyprus sit at the bottom, below Greece, Portugal and Bulgaria. Within-country spreads are wide almost everywhere except at the extremes. All of Finland's cities are in the top 61, and all of Romania's are below 245th. The gradient is consistent with the account of how high-cycling countries got there: separate facilities on heavily trafficked roads, traffic calming on residential streets, and restrictive policies on car use (Pucher & Buehler, 2008). It also echoes the widening gap in per-km cyclist fatality rates between the United States and the leading European cycling countries (Buehler & Pucher, 2020).

Two cautions apply. The gradient is partly a frame effect: Romania contributes twenty cities because the project began there. And it is partly a measurement of law. Statutory speed defaults lend every untagged street a limit, so a country's default speed law moves all its cities together, as in Spain. Neither caution reverses the finding. Romanian cities rank low on all three dimensions, not on one: their median ranks are 288 on street risk, 281 on journey risk and 270 on access, out of 300. Even the best Romanian city on each dimension ranks no higher than 172nd. The gradient is also not fixed. Cities can change their networks quickly: in 2020, pop-up lanes averaging 11.5 km per city raised cycling by 11 to 48% across 106 European cities within four months (Kraus & Koch, 2021). Research from south-eastern Europe adds that physical change may not be sufficient without attention to attitudes (Pojani et al., 2017). A measured gap is a gap a city can close, and because the index is computed identically every time, it can show whether a city is closing it.

### 5.6 Reading the index responsibly

Three rules follow from the results. Read tiers, not places: the median rank interval is 35 places wide, and only the extremes are tight. Read within-country comparisons with caution, because the criterion does not reach that far. And read index values as relative: the goalposts come from these 300 cities, so a different set of cities would move every value even if no street changed.

## 6. Limitations

**The road-level validation is in-sample.** The counter-panel re-run (Section 3.2) uses the same sites that helped audit the model's factors. It shows that the current scores are consistent with that panel, not that they generalise to other places. The city-level criterion covers only eighteen cities in five countries.

**The criterion is not fully independent.** The counter sites used for the city-level test come from the same counter programme that audited the road-level factors. A component that failed the audit was removed, so the model has, in a limited sense, seen this outcome data before. A clean test needs counter and injury data from cities outside that programme.

**Dimension weights are a design choice.** The weights of 0.40, 0.30 and 0.30 are judgements, as the weights of the indices in Table 1 are. The respecification analysis shows how much they matter. What distinguishes this index is that its output is checked against outcomes, not that its weights are free of judgement.

**The authors build the model.** The road-level model is developed by the operator of the routing service that publishes this index. Its factor magnitudes are proprietary. The ranking can be reproduced from the published city-level outputs, but not from first principles without the model.

**Country effects are not separated from city effects.** Section 5.3 discusses this. The outcome data cannot tell national road systems apart from national reporting.

**Junctions are under-measured.** The road-level signal survives control for road class mid-block but not at junctions. Junctions are where the injury literature places a large share of risk (Adams & Aldred, 2020) (Kapousizis et al., 2021).

**Traffic data are uneven.** Cities with measured counts are measured more accurately than cities without them. The model-only sensitivity shows the ranking is robust overall, but individual large cities move by up to 25 places.

**Map data are uneven.** OpenStreetMap completeness is high in Europe (Barrington-Leigh & Millard-Ball, 2017), but tagging practice varies. Statutory speed defaults fill gaps uniformly within a country and therefore move whole countries together.

**The frame is uneven.** Romania is over-represented, the units are administrative and of very different sizes, and the Canary Islands are outside coverage. The goalposts depend on the frame, so index values are relative to these 300 cities.

**Some things are not measured at all.** These include weather, bicycle parking, theft, culture, and driver behaviour beyond what the street encodes.

**Limits of the literature base.** The literature was retrieved through Crossref and OpenAlex. Of the 54 works cited, abstracts were read for 47; no full texts were read. Seven were available only as metadata and are cited for their existence and topic only, never for a finding: (Lowry et al., 2016), (Wang et al., 2016), (Harvey et al., 2024), (Winters et al., 2016), (Gitelman et al., 2010), (Wegman & Oppe, 2010) and (Lord & Mannering, 2010). Every printed DOI resolved at Crossref or DataCite. A resolved DOI shows that a work exists, not that it supports the sentence citing it, and each cited claim was checked against its abstract rather than the full text. The Copenhagenize Index and the Eurostat tables have no DOI and are named in the text without verification.

## 7. Conclusion

This paper ranked 300 European cities by the risk of the streets cyclists actually ride. Every city is measured with one road-level model that a routing service uses daily, aggregated into street, journey and access dimensions. The ranking is stable at the level that matters. Three tiers hold for 246 of the 300 cities under 2,000 respecifications, and only the extremes are tight at the level of individual ranks.

The ranking reproduces the known cycling leaders once an administrative-boundary artefact is corrected: Copenhagen's municipal outline had cut its bridges out of the network. It also finds leaders that reputation misses, notably Helsinki's suburbs. In the eighteen cities with exposure data, it predicts serious-injury rates per passage (rate ratio 0.78 per 10 index points), and it correlates moderately with cycling's share of commuting ($\rho = 0.539$). The injury association disappears when cities are compared only within their own country. That is the paper's main open question, and the honest scope of the result is between groups of cities, not between neighbours a few places apart.

Three things would strengthen future work: - counter and injury data from more countries and more cities per country, so that within-country validity can be tested; - road-level validation on counter and injury data from regions not used to audit the model; - better junction measurement, the largest known gap in the road-level model.

The method itself is designed to repeat. Because it is computed the same way every time, repeating it can show whether a city's streets are getting safer, which a ranking of reputations cannot.

## Data sources

Street network and city boundaries: © OpenStreetMap contributors, available under the Open Database Licence. Elevation: GEDTM30 (CC BY 4.0). Population: WorldPop (CC BY 4.0). Cycling mode share: Eurostat Urban Audit. Motor-traffic volumes combine published counts from public road and statistics authorities with a modelled estimate where no counts exist.

Validation counts come from public counter networks (MobiData BW for Baden-Württemberg, the Berlin Senate, the Canton of Basel-Stadt, the City of Zurich, the City of Madrid, and the open-data portals of Paris, Bordeaux Métropole, Nantes Métropole and Toulouse Métropole) and from citizen-science counters: Telraam in Belgium and France, and the Berlin zählt network of ADFC Berlin. Telraam data are licensed under CC BY-NC 4.0 and are used here with Telraam's permission. Injury records come from Statbel (Belgium), the German accident atlas (Unfallatlas) of the Federal and State Statistical Offices, the French national road-crash file (BAAC), the City of Zurich and the Swiss Federal Roads Office, and the City of Madrid.

Factor weights of the road-level model are proprietary. The paper names each factor and the direction of its effect but not its magnitude.

## References


Adams, T., & Aldred, R. (2020). Cycling Injury Risk in London: Impacts of Road Characteristics and Infrastructure. *Findings*. https://doi.org/10.32866/001c.18226

Aldred, R., Goel, R., Woodcock, J., & Goodman, A. (2017). Contextualising Safety in Numbers: a longitudinal investigation into change in cycling safety in Britain, 1991–2001 and 2001–2011. *Injury Prevention*. https://doi.org/10.1136/injuryprev-2017-042498

Aldred, R., Kapousizis, G., & Goodman, A. (2021). Association of Infrastructure and Route Environment Factors with Cycling Injury Risk at Intersection and Non-Intersection Locations: A Case-Crossover Study of Britain. *International Journal of Environmental Research and Public Health*. https://doi.org/10.3390/ijerph18063060

Barrington-Leigh, C., & Millard-Ball, A. (2017). The world's user-generated road map is more than 80% complete. *PLOS ONE*. https://doi.org/10.1371/journal.pone.0180698

Buehler, R., & Dill, J. (2015). Bikeway Networks: A Review of Effects on Cycling. *Transport Reviews*. https://doi.org/10.1080/01441647.2015.1069908

Buehler, R., & Pucher, J. (2020). The growing gap in pedestrian and cyclist fatality rates between the United States and the United Kingdom, Germany, Denmark, and the Netherlands, 1990–2018. *Transport Reviews*. https://doi.org/10.1080/01441647.2020.1823521

Chen, C., Lin, X., & Yu, G. (2026). Infrastructure matters: A systematic review of bicycle infrastructure design influencing cycling crash and injury risk. *Journal of Safety Research*. https://doi.org/10.1016/j.jsr.2026.06.008

Dadashova, B., Griffin, G. P., Das, S., Turner, S., & Sherman, B. (2020). Estimation of Average Annual Daily Bicycle Counts using Crowdsourced Strava Data. *Transportation Research Record: Journal of the Transportation Research Board*. https://doi.org/10.1177/0361198120946016

Dijkstra, L., Florczyk, A. J., Freire, S., Kemper, T., Melchiorri, M., & Pesaresi, M. (2021). Applying the Degree of Urbanisation to the globe: A new harmonised definition reveals a different picture of global urbanisation. *Journal of Urban Economics*. https://doi.org/10.1016/j.jue.2020.103312

Dill, J., & McNeil, N. (2016). Revisiting the Four Types of Cyclists: Findings from a National Survey. *Transportation Research Record: Journal of the Transportation Research Board*. https://doi.org/10.3141/2587-11

Elvik, R. (2009). The non-linearity of risk and the promotion of environmentally sustainable transport. *Accident Analysis & Prevention*. https://doi.org/10.1016/j.aap.2009.04.009

Ferenchak, N. N., & Marshall, W. E. (2020). Is bicycling getting safer? Bicycle fatality rates (1985–2017) using four exposure metrics. *Transportation Research Interdisciplinary Perspectives*. https://doi.org/10.1016/j.trip.2020.100219

Ferenchak, N. N., & Marshall, W. E. (2025). The link between low-stress bicycle facilities and bicycle commuting. *Nature Cities*. https://doi.org/10.1038/s44284-025-00255-5

Furth, P. G., Mekuria, M. C., & Nixon, H. (2016). Network Connectivity for Low-Stress Bicycling. *Transportation Research Record: Journal of the Transportation Research Board*. https://doi.org/10.3141/2587-06

Furth, P. G., Putta, T. V., & Moser, P. (2018). Measuring low-stress connectivity in terms of bike-accessible jobs and potential bike-to-work trips: A case study evaluating alternative bike route alignments in northern Delaware. *Journal of Transport and Land Use*. https://doi.org/10.5198/jtlu.2018.1159

Gitelman, V., Doveh, E., & Hakkert, S. (2010). Designing a composite indicator for road safety. *Safety Science*. https://doi.org/10.1016/j.ssci.2010.01.011

Greco, S., Ishizaka, A., Tasiou, M., & Torrisi, G. (2018). On the Methodological Framework of Composite Indices: A Review of the Issues of Weighting, Aggregation, and Robustness. *Social Indicators Research*. https://doi.org/10.1007/s11205-017-1832-9

Haklay, M. (2010). How Good is Volunteered Geographical Information? A Comparative Study of OpenStreetMap and Ordnance Survey Datasets. *Environment and Planning B: Planning and Design*. https://doi.org/10.1068/b35097

Hardinghaus, M., Nieland, S., Lehne, M., & Weschke, J. (2021). More than Bike Lanes—A Multifactorial Index of Urban Bikeability. *Sustainability*. https://doi.org/10.3390/su132111584

Harvey, C., Rodríguez, D. A., & Fang, K. (2024). Comparing methods and data sources for classifying bicycle level of traffic stress: How well do their outcomes agree?. *Sustainable Cities and Society*. https://doi.org/10.1016/j.scs.2023.105151

Huber, S., Lindemann, P., & Schröter, B. (2024). Safety and bicycle route choice: To what extent do accident risk and perceived safety influence bicycle route choice?. *Transportation Engineering*.

https://doi.org/10.1016/j.treng.2024.100240

Jacobsen, P. L. (2003). Safety in numbers: more walkers and bicyclists, safer walking and bicycling. *Injury Prevention*. https://doi.org/10.1136/ip.9.3.205

Kapousizis, G., Goodman, A., & Aldred, R. (2021). Cycling injury risk in Britain: A case-crossover study of infrastructural and route environment correlates. *Accident Analysis & Prevention*. https://doi.org/10.1016/j.aap.2021.106063

Kellershohn, J., Dickler, S., & Jungbluth, C. (2025). Model-Based Bikeability Indexing for Inter-City Comparisons to Evaluate Infrastructure and Level of Service for Cyclists. *Future Transportation*. https://doi.org/10.3390/futuretransp5020064

Kraus, S., & Koch, N. (2021). Provisional COVID-19 infrastructure induces large, rapid increases in cycling. *Proceedings of the National Academy of Sciences*. https://doi.org/10.1073/pnas.2024399118

Krenn, P. J., Oja, P., & Titze, S. (2015). Development of a Bikeability Index to Assess the Bicycle-Friendliness of Urban Environments. *Open Journal of Civil Engineering*. https://doi.org/10.4236/ojce.2015.54045

Lord, D., & Mannering, F. (2010). The statistical analysis of crash-frequency data: A review and assessment of methodological alternatives. *Transportation Research Part A: Policy and Practice*. https://doi.org/10.1016/j.tra.2010.02.001

Lowry, M. B., Furth, P., & Hadden-Loh, T. (2016). Prioritizing new bicycle facilities to improve low-stress network connectivity. *Transportation Research Part A: Policy and Practice*. https://doi.org/10.1016/j.tra.2016.02.003

Lubbe, N., Wu, Y., & Jeppsson, H. (2022). Safe speeds: fatality and injury risks of pedestrians, cyclists, motorcyclists, and car drivers impacting the front of another passenger car as a function of closing speed and age. *Traffic Safety Research*. https://doi.org/10.55329/vfma7555

Lusk, A. C., Furth, P. G., Morency, P., Miranda-Moreno, L. F., Willett, W. C., & Dennerlein, J. T. (2011). Risk of injury for bicycling on cycle tracks versus in the street. *Injury Prevention*. https://doi.org/10.1136/ip.2010.028696

Luxen, D., & Vetter, C. (2011). Real-time routing with OpenStreetMap data. *Proceedings of the 19th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems*. https://doi.org/10.1145/2093973.2094062

Macpherson, A. K., Zagorski, B., Saskin, R., Howard, A. W., Harris, M. A., & Namin, S. (2024). Comparison of the number of pedestrian and cyclist injuries captured in police data compared with health service utilisation data in Toronto, Canada 2016–2021. *Injury Prevention*. https://doi.org/10.1136/ip-2023-044974

Marqués, R., & Hernández-Herrador, V. (2017). On the effect of networks of cycle-tracks on the risk of cycling. The case of Seville. *Accident Analysis & Prevention*. https://doi.org/10.1016/j.aap.2017.03.004

Marshall, W. E., & Ferenchak, N. N. (2019). Why cities with high bicycling rates are safer for all road users. *Journal of Transport & Health*. https://doi.org/10.1016/j.jth.2019.03.004

Mulvaney, C. A., Smith, S., Watson, M. C., Parkin, J., Coupland, C., & Miller, P. (2015). Cycling infrastructure for reducing cycling injuries in cyclists. *Cochrane Database of Systematic Reviews*. https://doi.org/10.1002/14651858.cd010415.pub2

Nardo, M., Saisana, M., Saltelli, A., Tarantola, S., Hoffmann, A., & Giovannini, E. (2005). Handbook on Constructing Composite Indicators. *OECD Statistics Working Papers*. Organisation for Economic Co-Operation and Development (OECD). https://doi.org/10.1787/533411815016

Pojani, D., Bakija, D., Shkreli, E., Corcoran, J., & Mateo-Babiano, I. (2017). Do Northwestern and Southeastern Europe Share a Common "Cycling Mindset"? Comparative Analysis of Beliefs toward Cycling in the Netherlands and the Balkans. *European Journal of Transport and Infrastructure Research*. https://doi.org/10.18757/ejtir.2017.17.1.3178

Pucher, J., & Buehler, R. (2008). Making Cycling Irresistible: Lessons from The Netherlands, Denmark and Germany. *Transport Reviews*. https://doi.org/10.1080/01441640701806612

Reynolds, C. C., Harris, M. A., Teschke, K., Cripton, P. A., & Winters, M. (2009). The impact of transportation infrastructure on bicycling injuries and crashes: a review of the literature. *Environmental Health*. https://doi.org/10.1186/1476-069x-8-47

Rosén, E., & Sander, U. (2009). Pedestrian fatality risk as a function of car impact speed. *Accident Analysis & Prevention*. https://doi.org/10.1016/j.aap.2009.02.002

Saisana, M., Saltelli, A., & Tarantola, S. (2005). Uncertainty and Sensitivity Analysis Techniques as Tools for the Quality Assessment of Composite Indicators. *Journal of the Royal Statistical Society Series A: Statistics in Society*. https://doi.org/10.1111/j.1467-985x.2005.00350.x

Schepers, P., Heinen, E., Methorst, R., & Wegman, F. (2013). Road safety and bicycle usage impacts of unbundling vehicular and cycle traffic in Dutch urban networks. *European Journal of Transport and Infrastructure Research*. https://doi.org/10.18757/ejtir.2013.13.3.3000

Schmid-Querg, J., Keler, A., & Grigoropoulos, G. (2021). The Munich Bikeability Index: A Practical Approach for Measuring Urban Bikeability. *Sustainability*. https://doi.org/10.3390/su13010428

Schön, P., Heinen, E., & Manum, B. (2024). A scoping review on cycling network connectivity and its effects on cycling. *Transport Reviews*. https://doi.org/10.1080/01441647.2024.2337880

Tatem, A. J. (2017). WorldPop, open data for spatial demography. *Scientific Data*. https://doi.org/10.1038/sdata.2017.4

Teschke, K., Harris, M. A., Reynolds, C. C. O., Winters, M., Babul, S., & Chipman, M. (2012). Route Infrastructure and the Risk of Injuries to Bicyclists: A Case-Crossover Study. *American Journal of Public Health*. https://doi.org/10.2105/ajph.2012.300762

Vale, D. S., Saraiva, M., & Pereira, M. (2015). Active accessibility: A review of operational measures of walking and cycling accessibility. *Journal of Transport and Land Use*. https://doi.org/10.5198/jtlu.2015.593

van Petegem, J. H., Schepers, P., & Wijlhuizen, G. J. (2020). The safety of physically separated cycle tracks compared to marked cycle lanes and mixed traffic conditions in Amsterdam. *European Journal of Transport and Infrastructure Research*. https://doi.org/10.18757/ejtir.2021.21.3.5283

Wang, H., Palm, M., Chen, C., Vogt, R., & Wang, Y. (2016). Does bicycle network level of traffic stress (LTS) explain bicycle travel behavior? Mixed results from an Oregon case study. *Journal of Transport Geography*. https://doi.org/10.1016/j.jtrangeo.2016.08.016

Wegman, F., & Oppe, S. (2010). Benchmarking road safety performances of countries. *Safety Science*. https://doi.org/10.1016/j.ssci.2010.02.003

Winters, M., Brauer, M., Setton, E. M., & Teschke, K. (2013). Mapping Bikeability: A Spatial Tool to Support Sustainable Travel. *Environment and Planning B: Planning and Design*. https://doi.org/10.1068/b38185

Winters, M., Teschke, K., Brauer, M., & Fuller, D. (2016). Bike Score®: Associations Between Urban Bikeability and Cycling Behavior in 24 Cities. *Journal of Transport & Health*. https://doi.org/10.1016/j.jth.2016.05.021

Yannis, G., & Michelaraki, E. (2025). Effectiveness of 30 km/h speed limit – A literature review. *Journal of Safety Research*. https://doi.org/10.1016/j.jsr.2024.11.003

Łukawska, M., Chou, K., Paulsen, M., Rasmussen, T. K., & Nielsen, O. A. (2026). Bicycle route choice models with explicit safety indicators. *Transportation Research Part A: Policy and Practice*. https://doi.org/10.1016/j.tra.2026.105011

## Appendix A. Full ranking

**Table A1.** All 300 cities, ordered by index. The 90% rank interval and tier stability come from 2,000 respecifications; tier stability is the share of respecifications in which a city falls in its headline tier. S, R and A ranks run from 1 (lowest street risk, lowest journey risk, best low-stress access) to 300.

| Rank | City | Country | Index | 90% rank interval | Tier | Tier stability | S rank | R rank | A rank |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Stockholm | Sweden | 98.9 | 1–8 | 1 | 100% | 1 | 13 | 1 |
| 2 | Amsterdam | Netherlands | 97.9 | 2–15 | 1 | 100% | 9 | 9 | 2 |
| 3 | Helsinki | Finland | 97.9 | 1–11 | 1 | 100% | 7 | 1 | 15 |
| 4 | The Hague | Netherlands | 97.8 | 2–14 | 1 | 100% | 4 | 16 | 3 |
| 5 | Antwerp | Belgium | 97.7 | 3–15 | 1 | 100% | 10 | 10 | 9 |
| 6 | Vantaa | Finland | 97.2 | 2–11 | 1 | 100% | 12 | 2 | 11 |
| 7 | Leiden | Netherlands | 97.1 | 2–12 | 1 | 100% | 2 | 29 | 4 |
| 8 | Haarlem | Netherlands | 96.6 | 1–14 | 1 | 100% | 3 | 35 | 5 |
| 9 | Espoo | Finland | 95.5 | 3–16 | 1 | 100% | 13 | 3 | 22 |
| 10 | Copenhagen | Denmark | 95.0 | 4–22 | 1 | 100% | 6 | 24 | 17 |
| 11 | Tampere | Finland | 94.1 | 4–18 | 1 | 100% | 16 | 4 | 27 |
| 12 | Eindhoven | Netherlands | 93.8 | 10–26 | 1 | 100% | 11 | 25 | 21 |
| 13 | Paris | France | 93.7 | 6–19 | 1 | 100% | 17 | 33 | 6 |
| 14 | Utrecht | Netherlands | 93.5 | 10–24 | 1 | 100% | 15 | 17 | 19 |
| 15 | Rotterdam | Netherlands | 93.2 | 9–23 | 1 | 100% | 22 | 14 | 20 |
| 16 | Barcelona | Spain | 93.1 | 8–22 | 1 | 100% | 23 | 30 | 7 |
| 17 | Brussels | Belgium | 91.6 | 3–26 | 1 | 100% | 5 | 104 | 8 |
| 18 | Nijmegen | Netherlands | 90.5 | 15–30 | 1 | 100% | 8 | 28 | 44 |

| Rank | City | Country | Index | 90% rank interval | Tier | Tier stability | S rank | R rank | A rank |
|---|---|---|---|---|---|---|---|---|---|
| 19 | San Sebastián | Spain | 90.3 | 18–31 | 1 | 100% | 20 | 38 | 25 |
| 20 | Turku | Finland | 89.6 | 6–26 | 1 | 100% | 31 | 5 | 42 |
| 21 | Hanover | Germany | 89.0 | 19–32 | 1 | 100% | 27 | 27 | 28 |
| 22 | Arnhem | Netherlands | 88.0 | 19–36 | 1 | 100% | 18 | 22 | 48 |
| 23 | Munich | Germany | 88.0 | 21–36 | 1 | 100% | 24 | 49 | 26 |
| 24 | Ghent | Belgium | 86.7 | 23–41 | 1 | 100% | 19 | 58 | 35 |
| 25 | Bremen | Germany | 86.6 | 20–41 | 1 | 100% | 14 | 67 | 38 |
| 26 | Lyon | France | 85.7 | 13–38 | 1 | 100% | 51 | 75 | 10 |
| 27 | Braunschweig | Germany | 85.2 | 25–42 | 1 | 100% | 32 | 39 | 46 |
| 28 | Lübeck | Germany | 85.0 | 23–44 | 1 | 100% | 25 | 26 | 58 |
| 29 | Almere | Netherlands | 84.6 | 14–43 | 1 | 100% | 50 | 11 | 51 |
| 30 | Bilbao | Spain | 83.9 | 15–54 | 1 | 100% | 26 | 156 | 12 |
| 31 | Montpellier | France | 83.6 | 27–46 | 1 | 100% | 49 | 73 | 24 |
| 32 | Groningen | Netherlands | 83.5 | 28–43 | 1 | 100% | 55 | 40 | 34 |
| 33 | Toulouse | France | 83.0 | 29–50 | 1 | 100% | 29 | 95 | 39 |
| 34 | Albacete | Spain | 82.7 | 32–48 | 1 | 100% | 53 | 59 | 30 |
| 35 | Grenoble | France | 82.7 | 23–48 | 1 | 100% | 56 | 82 | 18 |
| 36 | Salamanca | Spain | 81.9 | 27–50 | 1 | 100% | 66 | 68 | 23 |
| 37 | Oldenburg | Germany | 81.6 | 33–54 | 1 | 100% | 34 | 100 | 40 |
| 38 | Strasbourg | France | 81.3 | 16–59 | 1 | 100% | 138 | 23 | 13 |
| 39 | Tilburg | Netherlands | 81.0 | 32–53 | 1 | 100% | 87 | 34 | 31 |
| 40 | Nantes | France | 80.6 | 29–61 | 1 | 100% | 21 | 117 | 55 |
| 41 | Las Rozas de Madrid | Spain | 80.1 | 37–55 | 1 | 100% | 37 | 71 | 52 |
| 42 | Málaga | Spain | 79.5 | 33–60 | 1 | 100% | 30 | 103 | 56 |
| 43 | Malmö | Sweden | 79.1 | 15–65 | 1 | 100% | 136 | 12 | 36 |
| 44 | Münster | Germany | 79.1 | 41–59 | 1 | 100% | 48 | 81 | 49 |
| 45 | Bologna | Italy | 78.8 | 28–58 | 1 | 100% | 79 | 19 | 60 |
| 46 | Seville | Spain | 78.5 | 32–67 | 1 | 100% | 132 | 21 | 33 |
| 47 | Valencia | Spain | 77.8 | 18–83 | 1 | 100% | 176 | 36 | 14 |
| 48 | Freiburg | Germany | 77.7 | 42–66 | 1 | 100% | 39 | 109 | 54 |
| 49 | Tours | France | 77.7 | 36–77 | 1 | 100% | 33 | 167 | 41 |
| 50 | Valladolid | Spain | 77.3 | 42–69 | 1 | 100% | 119 | 47 | 32 |
| 51 | Sabadell | Spain | 77.0 | 24–77 | 1 | 100% | 73 | 152 | 16 |
| 52 | Vitoria-Gasteiz | Spain | 76.2 | 46–67 | 1 | 100% | 95 | 43 | 53 |
| 53 | Duisburg | Germany | 76.0 | 43–72 | 1 | 100% | 35 | 126 | 65 |
| 54 | Pamplona | Spain | 75.9 | 46–74 | 1 | 100% | 115 | 60 | 37 |
| 55 | Burgos | Spain | 75.5 | 48–78 | 1 | 100% | 120 | 53 | 43 |
| 56 | Palma | Spain | 75.3 | 51–67 | 1 | 100% | 72 | 64 | 66 |
| 57 | Lille | France | 74.8 | 51–79 | 1 | 100% | 64 | 130 | 47 |
| 58 | Mannheim | Germany | 74.6 | 45–84 | 1 | 100% | 105 | 116 | 29 |
| 59 | Vienna | Austria | 74.6 | 42–88 | 1 | 100% | 155 | 31 | 45 |
| 60 | Zaragoza | Spain | 73.7 | 60–75 | 1 | 100% | 75 | 93 | 63 |

| Rank | City | Country | Index | 90% rank interval | Tier | Tier stability | S rank | R rank | A rank |
|---|---|---|---|---|---|---|---|---|---|
| 61 | Oulu | Finland | 73.2 | 12–86 | 1 | 99% | 146 | 6 | 83 |
| 62 | Córdoba | Spain | 73.1 | 49–82 | 1 | 100% | 46 | 134 | 73 |
| 63 | Nancy | France | 72.7 | 39–116 | 1 | 97% | 28 | 224 | 50 |
| 64 | Berlin | Germany | 72.4 | 50–83 | 1 | 100% | 44 | 119 | 89 |
| 65 | Sant Cugat del Vallès | Spain | 72.1 | 51–100 | 1 | 100% | 47 | 170 | 67 |
| 66 | Bonn | Germany | 71.9 | 49–102 | 1 | 100% | 38 | 181 | 69 |
| 67 | Augsburg | Germany | 71.8 | 58–82 | 1 | 100% | 58 | 112 | 79 |
| 68 | Magdeburg | Germany | 71.8 | 64–84 | 1 | 100% | 96 | 89 | 68 |
| 69 | Trondheim | Norway | 71.3 | 53–86 | 1 | 98% | 67 | 48 | 115 |
| 70 | Apeldoorn | Netherlands | 71.2 | 55–100 | 1 | 99% | 156 | 45 | 57 |
| 71 | Milton Keynes | United Kingdom | 70.7 | 12–98 | 1 | 95% | 124 | 7 | 124 |
| 72 | Düsseldorf | Germany | 70.7 | 66–96 | 1 | 100% | 69 | 139 | 70 |
| 73 | Graz | Austria | 70.6 | 65–89 | 1 | 100% | 102 | 62 | 80 |
| 74 | Oslo | Norway | 70.6 | 46–98 | 1 | 94% | 36 | 84 | 143 |
| 75 | Breda | Netherlands | 70.6 | 59–101 | 1 | 98% | 151 | 46 | 62 |
| 76 | Odense | Denmark | 70.4 | 41–96 | 1 | 98% | 129 | 20 | 99 |
| 77 | Maribor | Slovenia | 70.3 | 57–91 | 1 | 96% | 62 | 66 | 122 |
| 78 | Kiel | Germany | 70.3 | 69–93 | 1 | 100% | 80 | 124 | 71 |
| 79 | Gijón | Spain | 69.4 | 55–99 | 1 | 95% | 43 | 120 | 116 |
| 80 | Elche | Spain | 69.4 | 74–90 | 1 | 99% | 78 | 110 | 87 |
| 81 | Szeged | Hungary | 69.1 | 73–93 | 1 | 99% | 71 | 115 | 93 |
| 82 | Alicante | Spain | 68.4 | 78–96 | 1 | 99% | 103 | 101 | 84 |
| 83 | Rennes | France | 68.3 | 65–115 | 1 | 96% | 54 | 173 | 86 |
| 84 | Reggio Emilia | Italy | 68.2 | 13–129 | 1 | 88% | 215 | 8 | 77 |
| 85 | Aarhus | Denmark | 68.1 | 52–106 | 1 | 91% | 110 | 32 | 131 |
| 86 | Oberhausen | Germany | 67.9 | 57–125 | 1 | 91% | 42 | 188 | 96 |
| 87 | Murcia | Spain | 67.9 | 69–104 | 1 | 94% | 63 | 132 | 111 |
| 88 | Cologne | Germany | 67.1 | 74–118 | 1 | 98% | 117 | 148 | 64 |
| 89 | A Coruña | Spain | 67.1 | 72–116 | 1 | 96% | 139 | 127 | 61 |
| 90 | Santander | Spain | 66.7 | 87–102 | 1 | 94% | 97 | 111 | 100 |
| 91 | Białystok | Poland | 66.5 | 70–118 | 1 | 87% | 161 | 52 | 91 |
| 92 | Luxembourg | Luxembourg | 65.7 | 83–118 | 1 | 91% | 134 | 123 | 74 |
| 93 | Wrocław | Poland | 65.5 | 84–117 | 1 | 82% | 147 | 85 | 90 |
| 94 | Halle (Saale) | Germany | 65.4 | 79–120 | 1 | 79% | 65 | 149 | 123 |
| 95 | Karlsruhe | Germany | 65.1 | 86–133 | 1 | 83% | 86 | 174 | 85 |
| 96 | Ljubljana | Slovenia | 64.2 | 88–121 | 1 | 70% | 74 | 136 | 134 |
| 97 | Ludwigshafen am Rhein | Germany | 64.0 | 78–144 | 1 | 67% | 158 | 160 | 59 |
| 98 | Orléans | France | 63.5 | 98–120 | 1 | 61% | 121 | 122 | 107 |
| 99 | Castellón de la Plana | Spain | 63.1 | 81–153 | 1 | 60% | 199 | 72 | 76 |
| 100 | Uppsala | Sweden | 63.0 | 23–161 | 1 | 68% | 226 | 15 | 102 |
| 101 | Bergen | Norway | 62.8 | 80–139 | 2 | 38% | 59 | 141 | 162 |

| Rank | City | Country | Index | 90% rank interval | Tier | Tier stability | S rank | R rank | A rank |
|---|---|---|---|---|---|---|---|---|---|
| 102 | Leverkusen | Germany | 62.5 | 100–143 | 2 | 57% | 111 | 171 | 95 |
| 103 | Krefeld | Germany | 62.4 | 96–148 | 2 | 62% | 125 | 176 | 81 |
| 104 | Almería | Spain | 62.4 | 103–129 | 2 | 53% | 106 | 140 | 118 |
| 105 | Gelsenkirchen | Germany | 62.3 | 99–149 | 2 | 61% | 100 | 183 | 97 |
| 106 | Hamburg | Germany | 62.2 | 84–154 | 2 | 50% | 60 | 200 | 125 |
| 107 | Vigo | Spain | 62.2 | 81–161 | 2 | 50% | 57 | 213 | 113 |
| 108 | Gdańsk | Poland | 61.8 | 106–132 | 2 | 60% | 107 | 142 | 128 |
| 109 | Terrassa | Spain | 61.8 | 94–152 | 2 | 71% | 153 | 162 | 75 |
| 110 | Stavanger | Norway | 61.7 | 91–140 | 2 | 46% | 101 | 86 | 163 |
| 111 | Hamm | Germany | 61.4 | 99–154 | 2 | 79% | 142 | 175 | 78 |
| 112 | Mainz | Germany | 61.2 | 103–159 | 2 | 77% | 113 | 191 | 92 |
| 113 | Nuremberg | Germany | 61.2 | 111–145 | 2 | 79% | 122 | 157 | 109 |
| 114 | Rostock | Germany | 60.9 | 89–156 | 2 | 57% | 61 | 192 | 149 |
| 115 | Angers | France | 60.9 | 79–170 | 2 | 55% | 52 | 221 | 136 |
| 116 | Debrecen | Hungary | 60.9 | 93–143 | 2 | 60% | 162 | 69 | 139 |
| 117 | Toruń | Poland | 60.7 | 102–156 | 2 | 74% | 185 | 113 | 98 |
| 118 | Łódź | Poland | 60.6 | 109–135 | 2 | 68% | 128 | 114 | 141 |
| 119 | Alkmaar | Netherlands | 60.5 | 94–169 | 2 | 65% | 212 | 97 | 82 |
| 120 | Leipzig | Germany | 60.1 | 94–153 | 2 | 59% | 68 | 158 | 168 |
| 121 | Clermont-Ferrand | France | 60.1 | 110–155 | 2 | 93% | 149 | 159 | 101 |
| 122 | Jerez de la Frontera | Spain | 60.0 | 115–140 | 2 | 76% | 116 | 137 | 144 |
| 123 | Tallinn | Estonia | 59.8 | 69–157 | 2 | 51% | 166 | 41 | 178 |
| 124 | Granada | Spain | 59.8 | 103–175 | 2 | 76% | 81 | 216 | 112 |
| 125 | Budapest | Hungary | 59.6 | 103–151 | 2 | 71% | 167 | 80 | 146 |
| 126 | Potsdam | Germany | 59.3 | 99–177 | 2 | 73% | 70 | 218 | 132 |
| 127 | Linz | Austria | 59.0 | 101–175 | 2 | 75% | 214 | 92 | 103 |
| 128 | Bordeaux | France | 58.8 | 114–170 | 2 | 87% | 92 | 199 | 130 |
| 129 | Bern | Switzerland | 58.8 | 75–198 | 2 | 58% | 45 | 253 | 119 |
| 130 | Aachen | Germany | 58.7 | 103–182 | 2 | 79% | 76 | 223 | 121 |
| 131 | Warsaw | Poland | 58.7 | 83–172 | 2 | 66% | 203 | 50 | 138 |
| 132 | Liège | Belgium | 58.6 | 91–176 | 2 | 70% | 213 | 61 | 120 |
| 133 | Geneva | Switzerland | 58.5 | 108–166 | 2 | 82% | 83 | 189 | 152 |
| 134 | Venice | Italy | 58.5 | 66–174 | 2 | 57% | 202 | 37 | 156 |
| 135 | Bedford | United Kingdom | 58.4 | 90–175 | 2 | 50% | 84 | 83 | 228 |
| 136 | Mönchengladbach | Germany | 58.3 | 114–175 | 2 | 100% | 160 | 187 | 94 |
| 137 | Gdynia | Poland | 58.2 | 130–160 | 2 | 98% | 133 | 164 | 127 |
| 138 | Edinburgh | United Kingdom | 58.2 | 88–163 | 2 | 63% | 159 | 56 | 184 |
| 139 | Madrid | Spain | 58.2 | 110–161 | 2 | 79% | 175 | 90 | 153 |
| 140 | Reykjavík | Iceland | 57.8 | 115–177 | 2 | 94% | 197 | 131 | 104 |
| 141 | Mülheim an der Ruhr | Germany | 57.7 | 126–177 | 2 | 97% | 118 | 197 | 117 |
| 142 | Frankfurt | Germany | 57.6 | 123–173 | 2 | 100% | 152 | 182 | 108 |

| Rank | City | Country | Index | 90% rank interval | Tier | Tier stability | S rank | R rank | A rank |
|---|---|---|---|---|---|---|---|---|---|
| 143 | Valletta | Malta | 57.4 | 75–210 | 2 | 57% | 40 | 265 | 129 |
| 144 | Peterborough | United Kingdom | 57.2 | 85–174 | 2 | 66% | 182 | 51 | 182 |
| 145 | Aalborg | Denmark | 57.0 | 98–171 | 2 | 69% | 165 | 65 | 189 |
| 146 | Manchester | United Kingdom | 57.0 | 117–166 | 2 | 82% | 171 | 96 | 161 |
| 147 | York | United Kingdom | 56.9 | 112–171 | 2 | 72% | 130 | 98 | 196 |
| 148 | Oviedo | Spain | 56.8 | 76–207 | 2 | 59% | 41 | 256 | 145 |
| 149 | Poznań | Poland | 56.5 | 130–177 | 2 | 98% | 191 | 146 | 114 |
| 150 | Florence | Italy | 56.4 | 114–172 | 2 | 81% | 174 | 87 | 169 |
| 151 | Lublin | Poland | 55.9 | 119–175 | 2 | 82% | 173 | 94 | 175 |
| 152 | Bristol | United Kingdom | 55.7 | 117–181 | 2 | 76% | 99 | 133 | 212 |
| 153 | Gothenburg | Sweden | 55.5 | 125–189 | 2 | 89% | 216 | 102 | 135 |
| 154 | Rouen | France | 55.2 | 110–212 | 2 | 96% | 164 | 245 | 72 |
| 155 | Dortmund | Germany | 55.1 | 124–186 | 2 | 91% | 91 | 198 | 172 |
| 156 | Kraków | Poland | 54.9 | 140–173 | 2 | 93% | 123 | 163 | 176 |
| 157 | Brighton and Hove | United Kingdom | 54.8 | 136–179 | 2 | 93% | 114 | 180 | 174 |
| 158 | Mulhouse | France | 54.6 | 144–191 | 2 | 99% | 131 | 206 | 140 |
| 159 | Dublin | Ireland | 54.4 | 124–187 | 2 | 87% | 88 | 194 | 188 |
| 160 | Modena | Italy | 54.3 | 75–219 | 2 | 57% | 268 | 44 | 88 |
| 161 | Bielefeld | Germany | 54.1 | 135–191 | 2 | 93% | 104 | 203 | 164 |
| 162 | Nottingham | United Kingdom | 54.0 | 150–187 | 2 | 99% | 186 | 161 | 137 |
| 163 | Stuttgart | Germany | 53.8 | 134–208 | 2 | 96% | 109 | 239 | 133 |
| 164 | Darmstadt | Germany | 53.4 | 129–217 | 2 | 93% | 127 | 249 | 106 |
| 165 | Bratislava | Slovakia | 53.3 | 160–180 | 2 | 92% | 170 | 150 | 170 |
| 166 | Rzeszów | Poland | 53.2 | 158–183 | 2 | 94% | 183 | 147 | 157 |
| 167 | Linköping | Sweden | 52.9 | 138–208 | 2 | 99% | 208 | 193 | 105 |
| 168 | Swindon | United Kingdom | 52.4 | 104–199 | 2 | 67% | 220 | 54 | 194 |
| 169 | Kassel | Germany | 52.3 | 126–223 | 2 | 84% | 93 | 260 | 126 |
| 170 | Zurich | Switzerland | 52.3 | 143–210 | 2 | 89% | 112 | 234 | 154 |
| 171 | Dresden | Germany | 52.2 | 132–207 | 2 | 83% | 89 | 230 | 180 |
| 172 | Reims | France | 52.2 | 157–202 | 2 | 94% | 145 | 219 | 150 |
| 173 | Metz | France | 52.0 | 154–207 | 2 | 92% | 137 | 225 | 151 |
| 174 | Bydgoszcz | Poland | 51.9 | 154–197 | 2 | 86% | 204 | 128 | 158 |
| 175 | Bergamo | Italy | 51.9 | 140–191 | 2 | 77% | 181 | 106 | 206 |
| 176 | Plzeň | Czechia | 51.9 | 109–199 | 2 | 66% | 211 | 63 | 204 |
| 177 | Toulon | France | 51.8 | 139–205 | 2 | 82% | 98 | 222 | 186 |
| 178 | Turin | Italy | 51.8 | 39–223 | 2 | 45% | 267 | 18 | 142 |
| 179 | Portsmouth | United Kingdom | 51.6 | 168–193 | 2 | 87% | 190 | 151 | 171 |

| Rank | City | Country | Index | 90% rank interval | Tier | Tier stability | S rank | R rank | A rank |
|---|---|---|---|---|---|---|---|---|---|
| 180 | Parma | Italy | 51.6 | 81–218 | 2 | 56% | 257 | 42 | 147 |
| 181 | Essen | Germany | 51.3 | 161–204 | 2 | 85% | 135 | 214 | 165 |
| 182 | Riga | Latvia | 51.2 | 162–192 | 2 | 80% | 168 | 143 | 203 |
| 183 | Telford | United Kingdom | 50.8 | 128–200 | 2 | 67% | 201 | 79 | 214 |
| 184 | Newcastle upon Tyne | United Kingdom | 50.4 | 153–197 | 2 | 73% | 187 | 118 | 213 |
| 185 | Bochum | Germany | 50.1 | 141–217 | 2 | 71% | 90 | 238 | 191 |
| 186 | Szczecin | Poland | 50.1 | 171–207 | 2 | 76% | 210 | 153 | 160 |
| 187 | Kielce | Poland | 50.0 | 180–203 | 2 | 77% | 194 | 165 | 166 |
| 188 | Basel | Switzerland | 49.9 | 149–223 | 2 | 81% | 192 | 236 | 110 |
| 189 | Le Havre | France | 49.7 | 173–201 | 2 | 73% | 150 | 177 | 205 |
| 190 | Leeds | United Kingdom | 49.2 | 132–213 | 2 | 59% | 234 | 76 | 192 |
| 191 | Birmingham | United Kingdom | 49.1 | 111–213 | 2 | 55% | 222 | 55 | 237 |
| 192 | Saint-Étienne | France | 49.0 | 150–228 | 2 | 63% | 108 | 254 | 159 |
| 193 | Swansea | United Kingdom | 48.7 | 127–213 | 2 | 58% | 209 | 70 | 241 |
| 194 | Cartagena | Spain | 48.5 | 166–220 | 2 | 60% | 126 | 237 | 183 |
| 195 | Verona | Italy | 48.4 | 139–214 | 2 | 57% | 232 | 91 | 198 |
| 196 | Pescara | Italy | 48.4 | 155–209 | 2 | 60% | 219 | 108 | 208 |
| 197 | Zagreb | Croatia | 48.4 | 172–212 | 2 | 64% | 143 | 178 | 233 |
| 198 | Hagen | Germany | 48.2 | 146–235 | 2 | 58% | 94 | 263 | 173 |
| 199 | London | United Kingdom | 48.2 | 181–207 | 2 | 62% | 180 | 155 | 221 |
| 200 | Lausanne | Switzerland | 47.9 | 130–239 | 2 | 56% | 77 | 268 | 185 |
| 201 | Alcalá de Henares | Spain | 47.9 | 167–230 | 3 | 46% | 140 | 250 | 155 |
| 202 | Prague | Czechia | 47.5 | 188–212 | 3 | 43% | 169 | 186 | 215 |
| 203 | Southampton | United Kingdom | 47.3 | 194–211 | 3 | 46% | 179 | 184 | 207 |
| 204 | Vilnius | Lithuania | 46.9 | 187–217 | 3 | 50% | 154 | 210 | 210 |
| 205 | Saarbrücken | Germany | 46.5 | 139–244 | 3 | 48% | 82 | 274 | 181 |
| 206 | Plymouth | United Kingdom | 46.1 | 182–219 | 3 | 50% | 193 | 144 | 243 |
| 207 | Wolverhampton | United Kingdom | 46.1 | 146–225 | 3 | 50% | 240 | 88 | 219 |
| 208 | Wuppertal | Germany | 45.9 | 182–233 | 3 | 64% | 144 | 246 | 187 |
| 209 | Derby | United Kingdom | 45.8 | 161–221 | 3 | 52% | 228 | 105 | 227 |
| 210 | Glasgow | United Kingdom | 45.8 | 141–227 | 3 | 50% | 239 | 78 | 226 |
| 211 | Brescia | Italy | 45.5 | 123–237 | 3 | 49% | 256 | 57 | 200 |
| 212 | Nice | France | 45.4 | 197–219 | 3 | 68% | 217 | 166 | 199 |
| 213 | Sunderland | United Kingdom | 44.6 | 164–230 | 3 | 56% | 223 | 107 | 257 |

| Rank | City | Country | Index | 90% rank interval | Tier | Tier stability | S rank | R rank | A rank |
|---|---|---|---|---|---|---|---|---|---|
| 214 | Sheffield | United Kingdom | 44.6 | 198–225 | 3 | 63% | 172 | 196 | 244 |
| 215 | Warrington | United Kingdom | 44.2 | 186–235 | 3 | 72% | 251 | 145 | 179 |
| 216 | Katowice | Poland | 43.9 | 205–225 | 3 | 74% | 184 | 202 | 234 |
| 217 | Częstochowa | Poland | 43.8 | 170–247 | 3 | 68% | 265 | 135 | 148 |
| 218 | Erfurt | Germany | 43.5 | 205–230 | 3 | 92% | 195 | 227 | 195 |
| 219 | Leicester | United Kingdom | 43.0 | 186–236 | 3 | 74% | 247 | 129 | 209 |
| 220 | Chemnitz | Germany | 42.4 | 199–239 | 3 | 83% | 163 | 240 | 229 |
| 221 | Radom | Poland | 42.3 | 202–237 | 3 | 87% | 246 | 168 | 190 |
| 222 | Marseille | France | 42.3 | 207–233 | 3 | 88% | 177 | 226 | 235 |
| 223 | Prato | Italy | 42.2 | 147–250 | 3 | 64% | 270 | 74 | 202 |
| 224 | Gliwice | Poland | 42.0 | 203–242 | 3 | 96% | 188 | 247 | 193 |
| 225 | Basingstoke | United Kingdom | 41.9 | 215–232 | 3 | 94% | 230 | 195 | 201 |
| 226 | Sosnowiec | Poland | 41.5 | 219–232 | 3 | 96% | 221 | 205 | 217 |
| 227 | Genoa | Italy | 40.1 | 196–252 | 3 | 86% | 141 | 262 | 238 |
| 228 | Fuenlabrada | Spain | 39.7 | 227–238 | 3 | 100% | 225 | 220 | 216 |
| 229 | Stoke-on-Trent | United Kingdom | 39.6 | 172–251 | 3 | 74% | 255 | 99 | 259 |
| 230 | Plovdiv | Bulgaria | 39.6 | 156–256 | 3 | 71% | 271 | 77 | 236 |
| 231 | Thessaloniki | Greece | 39.6 | 167–259 | 3 | 71% | 85 | 272 | 267 |
| 232 | Kingston upon Hull | United Kingdom | 39.5 | 206–243 | 3 | 88% | 245 | 154 | 242 |
| 233 | Kaunas | Lithuania | 39.5 | 225–240 | 3 | 99% | 236 | 209 | 211 |
| 234 | Košice | Slovakia | 39.5 | 218–242 | 3 | 99% | 198 | 233 | 231 |
| 235 | Ostrava | Czechia | 39.3 | 198–247 | 3 | 85% | 252 | 138 | 245 |
| 236 | Medway | United Kingdom | 39.0 | 218–245 | 3 | 93% | 206 | 190 | 275 |
| 237 | Cardiff | United Kingdom | 38.3 | 222–247 | 3 | 100% | 207 | 244 | 223 |
| 238 | Brno | Czechia | 38.1 | 232–244 | 3 | 100% | 231 | 228 | 220 |
| 239 | Wiesbaden | Germany | 37.7 | 216–257 | 3 | 100% | 227 | 261 | 177 |
| 240 | Coventry | United Kingdom | 37.3 | 196–257 | 3 | 85% | 260 | 121 | 256 |
| 241 | Colchester | United Kingdom | 37.0 | 231–251 | 3 | 100% | 250 | 207 | 218 |
| 242 | Milan | Italy | 36.8 | 218–251 | 3 | 95% | 244 | 169 | 260 |
| 243 | Bradford | United Kingdom | 36.7 | 229–250 | 3 | 98% | 224 | 201 | 273 |
| 244 | Padua | Italy | 36.5 | 234–252 | 3 | 100% | 249 | 208 | 224 |
| 245 | Varna | Bulgaria | 35.9 | 227–254 | 3 | 100% | 200 | 248 | 254 |
| 246 | Cluj-Napoca | Romania | 35.5 | 230–256 | 3 | 100% | 205 | 251 | 247 |
| 247 | Lisbon | Portugal | 33.9 | 203–269 | 3 | 91% | 282 | 125 | 232 |
| 248 | Charleroi | Belgium | 33.4 | 228–261 | 3 | 98% | 262 | 179 | 252 |

| Rank | City | Country | Index | 90% rank interval | Tier | Tier stability | S rank | R rank | A rank |
|---|---|---|---|---|---|---|---|---|---|
| 249 | Luton | United Kingdom | 33.3 | 248–256 | 3 | 100% | 241 | 235 | 251 |
| 250 | Cork | Ireland | 32.5 | 243–262 | 3 | 100% | 229 | 257 | 246 |
| 251 | Trieste | Italy | 32.4 | 248–258 | 3 | 100% | 243 | 229 | 264 |
| 252 | Dijon | France | 32.1 | 233–268 | 3 | 100% | 263 | 255 | 167 |
| 253 | Sofia | Bulgaria | 31.8 | 250–260 | 3 | 100% | 233 | 242 | 270 |
| 254 | Porto | Portugal | 31.6 | 224–268 | 3 | 100% | 178 | 278 | 265 |
| 255 | Rome | Italy | 31.2 | 243–266 | 3 | 100% | 235 | 215 | 293 |
| 256 | Northampton | United Kingdom | 30.8 | 248–263 | 3 | 100% | 254 | 217 | 277 |
| 257 | Sintra | Portugal | 30.3 | 244–267 | 3 | 100% | 218 | 252 | 289 |
| 258 | Aberdeen | United Kingdom | 30.1 | 252–266 | 3 | 100% | 248 | 259 | 240 |
| 259 | Belfast | United Kingdom | 29.8 | 248–266 | 3 | 100% | 261 | 212 | 276 |
| 260 | Cagliari | Italy | 29.5 | 245–271 | 3 | 100% | 277 | 204 | 250 |
| 261 | Patras | Greece | 28.7 | 217–279 | 3 | 99% | 157 | 293 | 248 |
| 262 | Liverpool | United Kingdom | 28.6 | 256–268 | 3 | 100% | 264 | 231 | 253 |
| 263 | Naples | Italy | 28.4 | 232–275 | 3 | 100% | 189 | 284 | 263 |
| 264 | Palermo | Italy | 28.3 | 258–270 | 3 | 100% | 242 | 243 | 287 |
| 265 | Messina | Italy | 26.8 | 221–280 | 3 | 99% | 148 | 288 | 292 |
| 266 | Arad | Romania | 26.4 | 260–273 | 3 | 100% | 269 | 232 | 271 |
| 267 | Satu Mare | Romania | 26.2 | 233–284 | 3 | 100% | 295 | 185 | 197 |
| 268 | Doncaster | United Kingdom | 25.9 | 264–274 | 3 | 100% | 258 | 241 | 282 |
| 269 | Taranto | Italy | 25.5 | 264–272 | 3 | 100% | 253 | 267 | 269 |
| 270 | Oradea | Romania | 25.5 | 239–284 | 3 | 99% | 287 | 172 | 255 |
| 271 | Nicosia | Cyprus | 25.3 | 267–272 | 3 | 100% | 259 | 258 | 266 |
| 272 | Târgu Mureș | Romania | 23.3 | 261–279 | 3 | 100% | 279 | 269 | 222 |
| 273 | Reggio Calabria | Italy | 23.1 | 245–286 | 3 | 100% | 196 | 292 | 280 |
| 274 | Burgas | Bulgaria | 22.0 | 258–286 | 3 | 100% | 285 | 211 | 286 |
| 275 | Brașov | Romania | 21.7 | 264–281 | 3 | 100% | 275 | 280 | 225 |
| 276 | Maidstone | United Kingdom | 21.3 | 262–284 | 3 | 100% | 238 | 287 | 278 |
| 277 | Sibiu | Romania | 20.7 | 273–279 | 3 | 100% | 274 | 275 | 258 |
| 278 | Bournemouth | United Kingdom | 20.4 | 271–281 | 3 | 100% | 273 | 266 | 284 |
| 279 | Bari | Italy | 19.6 | 272–285 | 3 | 100% | 276 | 264 | 288 |
| 280 | Braga | Portugal | 19.2 | 274–283 | 3 | 100% | 266 | 276 | 283 |
| 281 | Timișoara | Romania | 18.9 | 275–282 | 3 | 100% | 278 | 271 | 272 |
| 282 | Catania | Italy | 17.5 | 266–290 | 3 | 100% | 237 | 291 | 294 |
| 283 | Constanța | Romania | 16.6 | 280–286 | 3 | 100% | 281 | 279 | 274 |
| 284 | Athens | Greece | 16.2 | 281–285 | 3 | 100% | 280 | 281 | 261 |
| 285 | Baia Mare | Romania | 14.9 | 279–289 | 3 | 100% | 272 | 290 | 268 |

| Rank | City | Country | Index | 90% rank interval | Tier | Tier stability | S rank | R rank | A rank |
|---|---|---|---|---|---|---|---|---|---|
| 286 | Bucharest | Romania | 14.6 | 277–289 | 3 | 100% | 283 | 289 | 239 |
| 287 | Limassol | Cyprus | 14.1 | 279–290 | 3 | 100% | 286 | 270 | 290 |
| 288 | Split | Croatia | 12.3 | 278–294 | 3 | 100% | 284 | 294 | 230 |
| 289 | Bacău | Romania | 12.0 | 282–289 | 3 | 100% | 292 | 277 | 262 |
| 290 | Pitești | Romania | 9.5 | 287–293 | 3 | 100% | 294 | 285 | 249 |
| 291 | Strovolos | Cyprus | 8.8 | 285–294 | 3 | 100% | 293 | 273 | 295 |
| 292 | Craiova | Romania | 7.2 | 290–294 | 3 | 100% | 288 | 286 | 285 |
| 293 | Galați | Romania | 6.0 | 291–295 | 3 | 100% | 289 | 282 | 296 |
| 294 | Paphos | Cyprus | 5.3 | 292–295 | 3 | 100% | 290 | 283 | 297 |
| 295 | Botoșani | Romania | 2.6 | 292–296 | 3 | 100% | 296 | 295 | 279 |
| 296 | Buzău | Romania | 2.0 | 294–298 | 3 | 100% | 297 | 296 | 281 |
| 297 | Ploiești | Romania | 1.4 | 295–298 | 3 | 100% | 291 | 297 | 298 |
| 298 | Brăila | Romania | 0.6 | 297–299 | 3 | 100% | 298 | 298 | 291 |
| 299 | Iași | Romania | 0.0 | 299–300 | 3 | 100% | 299 | 299 | 299 |
| 300 | Larnaca | Cyprus | 0.0 | 296–300 | 3 | 100% | 300 | 300 | 300 |

# Appendix B. Countries

**Table B1.** All 31 countries, ordered by median city index.

| Country | Cities | Median index | Mean index | Best city (rank) | Lowest city (rank) | Tier 1 | Tier 2 | Tier 3 |
|---|---|---|---|---|---|---|---|---|
| Finland | 6 | 94.8 | 91.2 | Helsinki (3) | Oulu (61) | 6 | 0 | 0 |
| Netherlands | 15 | 90.5 | 86.7 | Amsterdam (2) | Alkmaar (119) | 14 | 1 | 0 |
| Belgium | 5 | 86.7 | 73.6 | Antwerp (5) | Charleroi (248) | 3 | 1 | 1 |
| Spain | 35 | 72.1 | 70.1 | Barcelona (16) | Fuenlabrada (228) | 25 | 8 | 2 |
| Austria | 3 | 70.6 | 68.1 | Vienna (59) | Linz (127) | 2 | 1 | 0 |
| Denmark | 4 | 69.2 | 72.6 | Copenhagen (10) | Aalborg (145) | 3 | 1 | 0 |
| Slovenia | 2 | 67.2 | 67.2 | Maribor (77) | Ljubljana (96) | 2 | 0 | 0 |
| Norway | 4 | 66.7 | 66.6 | Trondheim (69) | Stavanger (110) | 2 | 2 | 0 |
| Luxembourg | 1 | 65.7 | 65.7 | Luxembourg (92) | – | 1 | 0 | 0 |
| Sweden | 5 | 63.0 | 69.9 | Stockholm (1) | Linköping (167) | 3 | 2 | 0 |
| Germany | 49 | 62.2 | 63.5 | Hanover (21) | Wiesbaden (239) | 21 | 23 | 5 |
| Hungary | 3 | 60.9 | 63.2 | Szeged (81) | Budapest (125) | 1 | 2 | 0 |
| France | 25 | 60.9 | 64.5 | Paris (13) | Dijon (252) | 12 | 10 | 3 |
| Estonia | 1 | 59.8 | 59.8 | Tallinn (123) | – | 0 | 1 | 0 |
| Iceland | 1 | 57.8 | 57.8 | Reykjavík (140) | – | 0 | 1 | 0 |
| Malta | 1 | 57.4 | 57.4 | Valletta (143) | – | 0 | 1 | 0 |
| Poland | 19 | 54.9 | 53.6 | Białystok (91) | Sosnowiec (226) | 2 | 12 | 5 |
| Switzerland | 5 | 52.3 | 53.5 | Bern (129) | Lausanne (200) | 0 | 5 | 0 |
| Latvia | 1 | 51.2 | 51.2 | Riga (182) | – | 0 | 1 | 0 |
| Slovakia | 2 | 46.4 | 46.4 | Bratislava (165) | Košice (234) | 0 | 1 | 1 |
| United Kingdom | 42 | 45.8 | 44.3 | Milton Keynes (71) | Bournemouth (278) | 1 | 16 | 25 |
| Ireland | 2 | 43.5 | 43.5 | Dublin (159) | Cork (250) | 0 | 1 | 1 |

| Country | Cities | Median index | Mean index | Best city (rank) | Lowest city (rank) | Tier 1 | Tier 2 | Tier 3 |
|---|---|---|---|---|---|---|---|---|
| Czechia | 4 | 43.4 | 44.2 | Plzeň (176) | Brno (238) | 0 | 1 | 3 |
| Lithuania | 2 | 43.2 | 43.2 | Vilnius (204) | Kaunas (233) | 0 | 0 | 2 |
| Italy | 25 | 40.1 | 41.3 | Bologna (45) | Catania (282) | 2 | 8 | 15 |
| Bulgaria | 4 | 33.9 | 32.3 | Plovdiv (230) | Burgas (274) | 0 | 0 | 4 |
| Portugal | 4 | 31.0 | 28.8 | Lisbon (247) | Braga (280) | 0 | 0 | 4 |
| Croatia | 2 | 30.4 | 30.4 | Zagreb (197) | Split (288) | 0 | 1 | 1 |
| Greece | 3 | 28.7 | 28.2 | Thessaloniki (231) | Athens (284) | 0 | 0 | 3 |
| Romania | 20 | 14.8 | 14.3 | Cluj-Napoca (246) | Iași (299) | 0 | 0 | 20 |
| Cyprus | 5 | 8.8 | 10.7 | Nicosia (271) | Larnaca (300) | 0 | 0 | 5 |
| **All** | **300** | **56.1** | **54.9** | | | **100** | **100** | **100** |

# Appendix C. Criterion-validity panel

**Table C1.** The 18 indexed cities with counter-measured exposure and police-recorded injuries.

| City | Country | Index | Tier | Counter sites | Passages (millions) | KSI | All crashes | KSI per 10M passages |
|---|---|---|---|---|---|---|---|---|
| Antwerp | Belgium | 97.7 | 1 | 26 | 51.9 | 11 | 252 | 2.12 |
| Paris | France | 93.7 | 1 | 62 | 160.3 | 31 | 621 | 1.93 |
| Brussels | Belgium | 91.6 | 1 | 31 | 51.6 | 6 | 186 | 1.16 |
| Ghent | Belgium | 86.7 | 1 | 13 | 63.5 | 10 | 60 | 1.57 |
| Toulouse | France | 83.0 | 1 | 77 | 106.1 | 19 | 56 | 1.79 |
| Nantes | France | 80.6 | 1 | 15 | 22.2 | 3 | 7 | 1.35 |
| Freiburg | Germany | 77.7 | 1 | 16 | 214.0 | 45 | 349 | 2.10 |
| Mannheim | Germany | 74.6 | 1 | 17 | 85.2 | 30 | 162 | 3.52 |
| Berlin | Germany | 72.4 | 1 | 189 | 815.6 | 381 | 2992 | 4.67 |
| Karlsruhe | Germany | 65.1 | 1 | 1 | 15.7 | 5 | 57 | 3.18 |
| Potsdam | Germany | 59.3 | 2 | 2 | 9.4 | 7 | 76 | 7.47 |
| Bordeaux | France | 58.8 | 2 | 28 | 42.4 | 17 | 37 | 4.01 |
| Liège | Belgium | 58.6 | 2 | 12 | 14.4 | 0 | 24 | 0.00 |
| Madrid | Spain | 58.2 | 2 | 134 | 76.8 | 26 | 705 | 3.39 |
| Stuttgart | Germany | 53.8 | 2 | 18 | 57.4 | 13 | 171 | 2.27 |
| Zurich | Switzerland | 52.3 | 2 | 25 | 212.2 | 157 | 826 | 7.40 |
| Basel | Switzerland | 49.9 | 2 | 30 | 169.7 | 96 | 274 | 5.66 |
| Charleroi | Belgium | 33.4 | 3 | 5 | 1.4 | 0 | 0 | 0.00 |
| **Total** | | | | **701** | **2,169.8** | **857** | **6855** | **3.95** |

# Appendix D. Figures

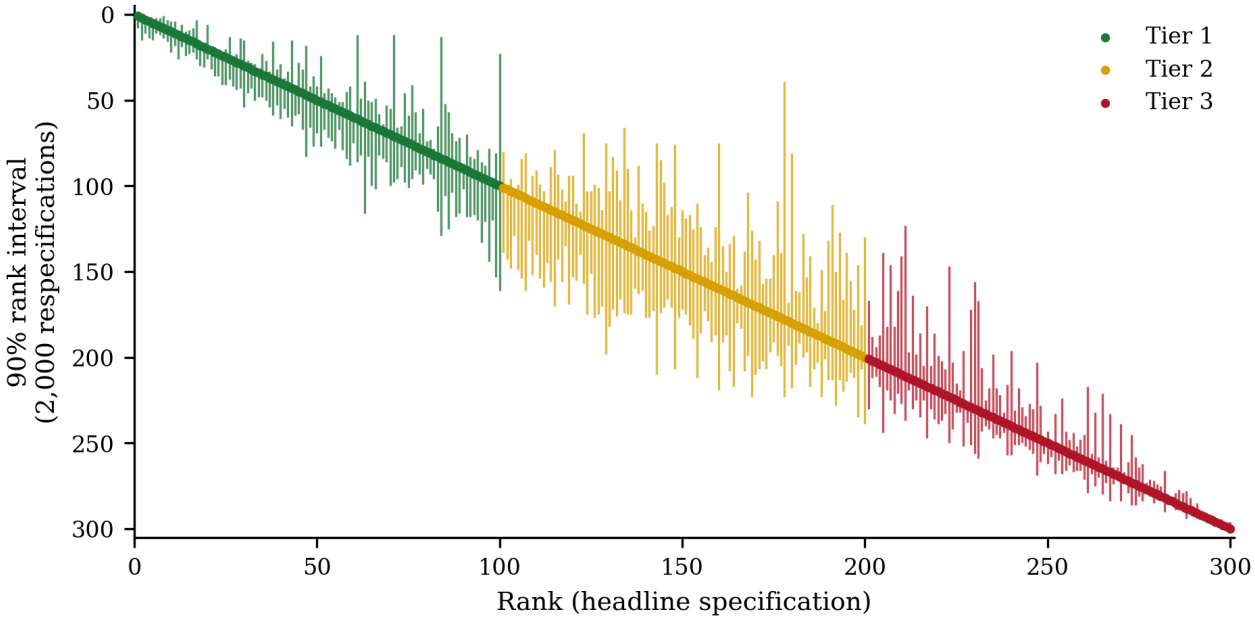



**Figure D1.** Headline rank and 90% rank interval from 2,000 respecifications for all 300 cities, coloured by tier.

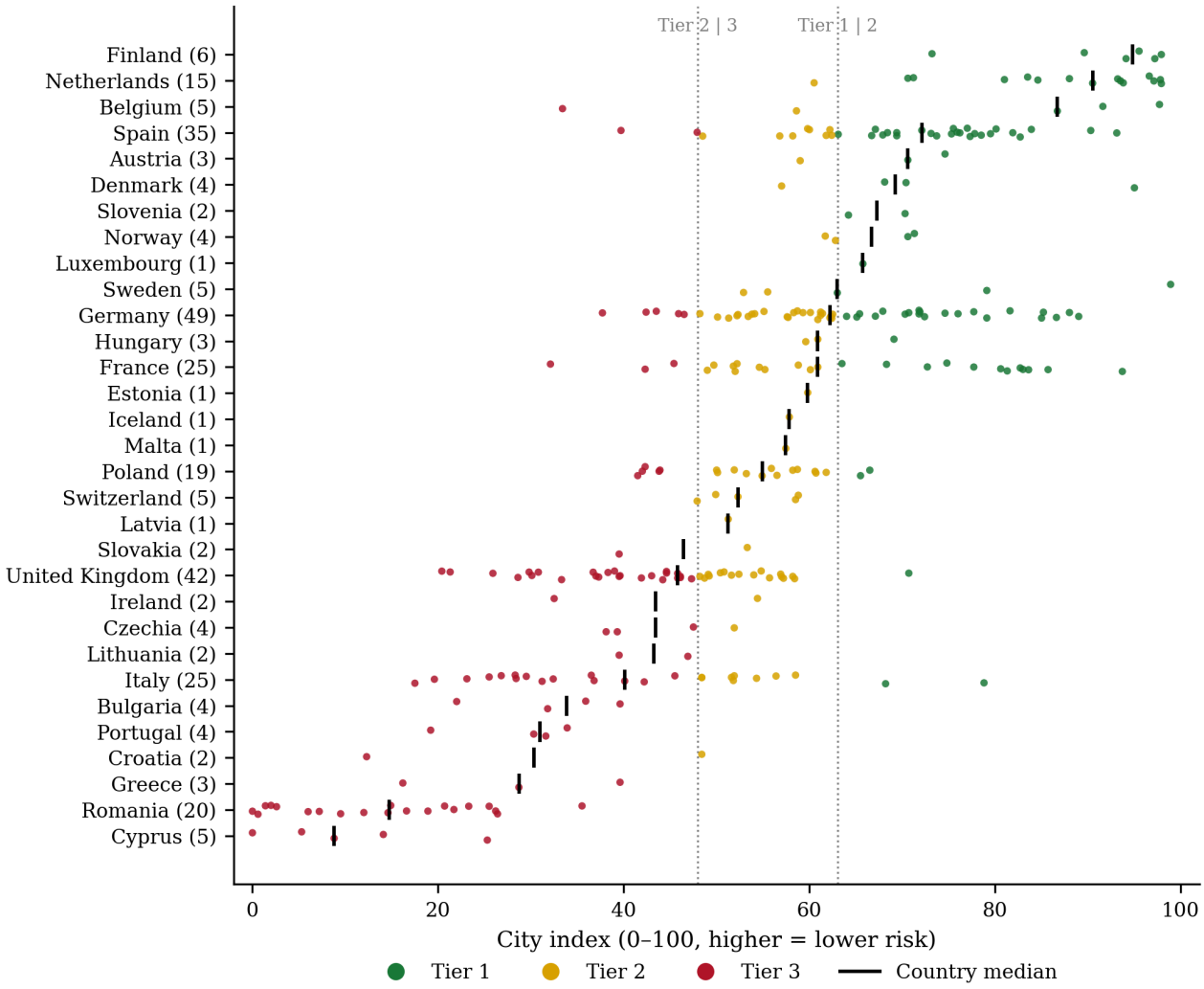


**Figure D2.** City index by country (number of indexed cities in brackets), with country medians and the two tier thresholds.

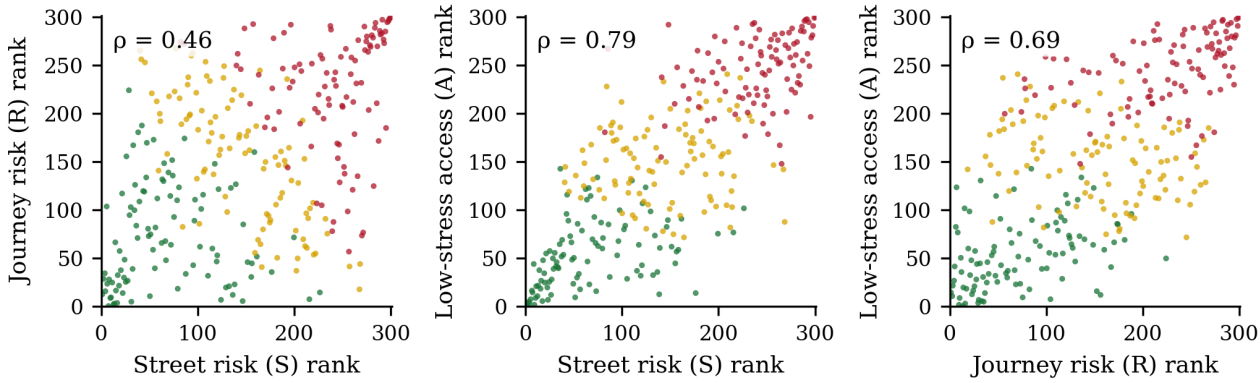



**Figure D3.** Rank relationships between the three dimensions: street risk (S), journey risk (R) and low-stress access (A), coloured by tier. ρ is Spearman's rank correlation.

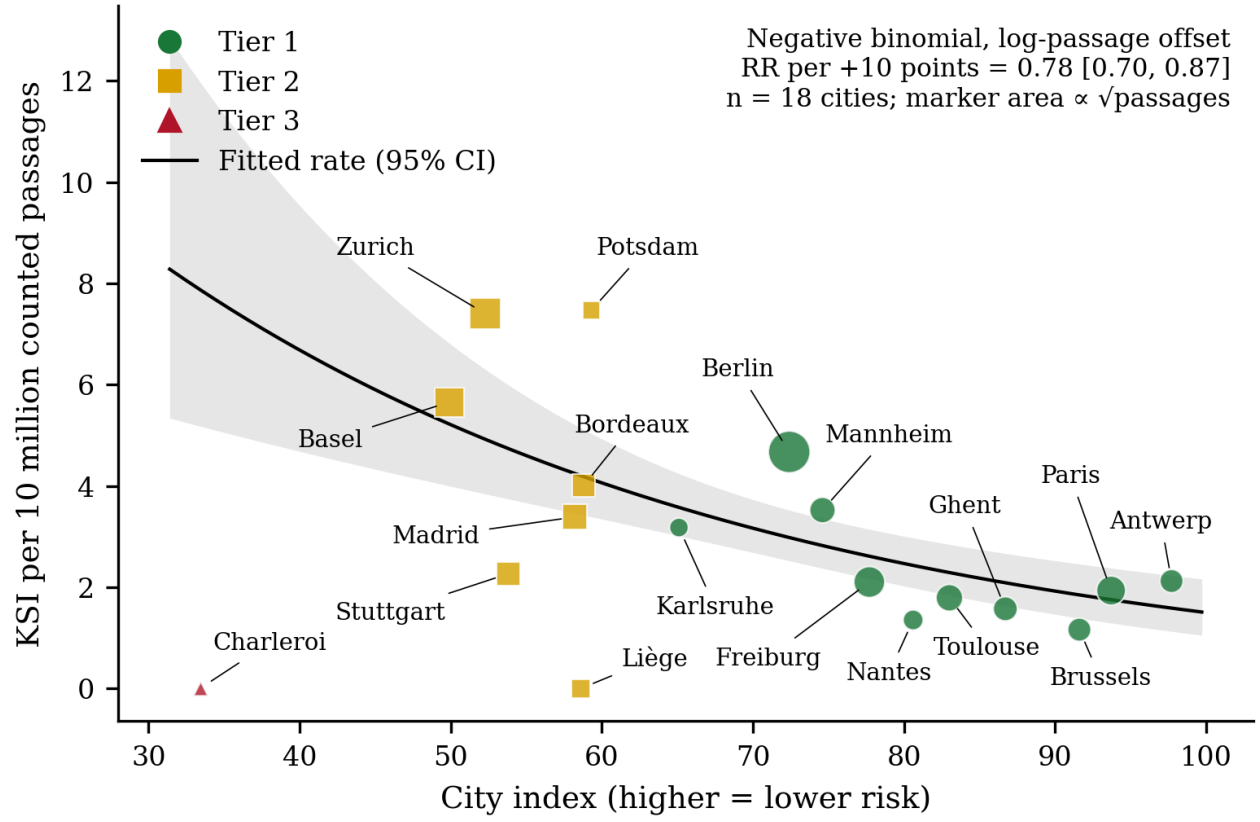



**Figure D4.** Serious injuries (KSI) per 10 million counted passages against the city index for the 18 cities in the criterion panel, with the fitted negative binomial rate and its 95% confidence band. Marker area is proportional to the square root of passages.

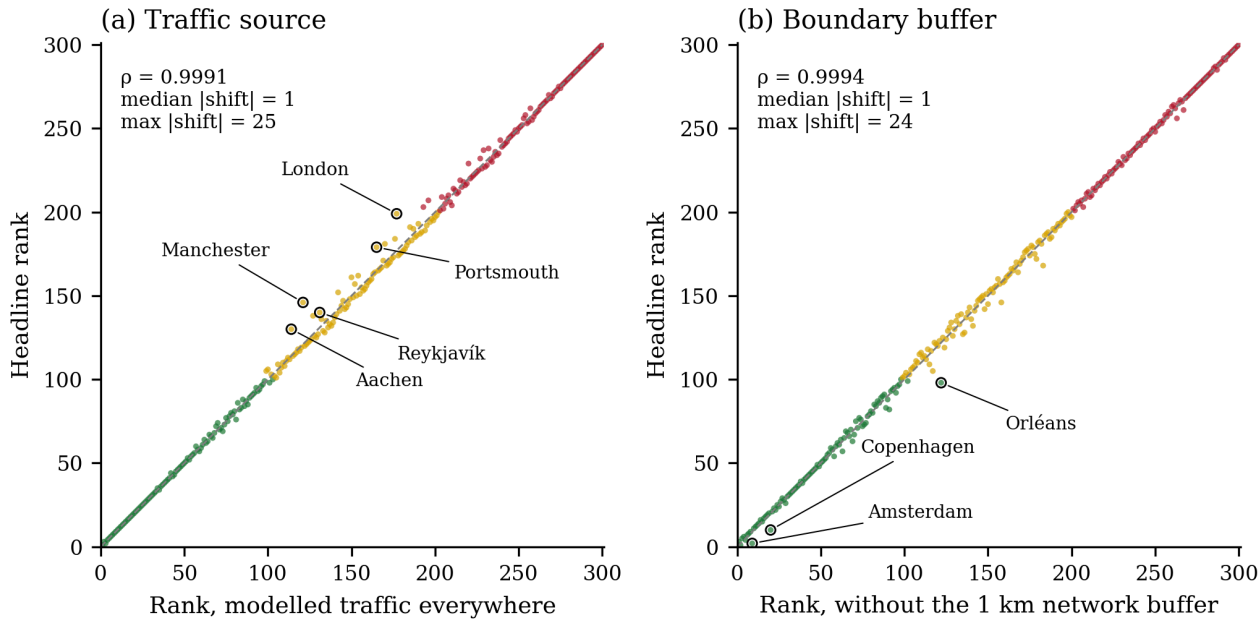



**Figure D5.** Sensitivity of the headline rank (a) to using modelled traffic everywhere and (b) to removing the 1 km network buffer. The labelled cities move most.